

Semisimple Deformations of Contraction Algebras for Simple Curve Flops

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Abstract

Let A be the contraction algebra of a simple flopping curve in a smooth complex threefold, with genus-zero Gopakumar–Vafa invariants n_j . We construct a finite free $\mathbb{C}[[s]]$ -deformation whose geometric generic fibre has n_j matrix blocks of size j , and prove that this is the unique semisimple deformation target of A . We also establish $\dim HH_0(A) = \dim Z(A) = \sum_j n_j$ and smoothness of A in the scheme of associative multiplications with fixed identity.

The existence proof combines a marked ADE morsification with the local matrix factors of a tilting generator. For uniqueness, we present the universal marked cocenter by a square matrix whose determinant defines the reduced curve discriminant. Restricting to the classifying arc computes the cocenter dimension; symmetry, bimodule periodicity, and Tate duality then give the required orbit-dimension equality. The coefficient-five E_8 case uses an exact modular calculation and a graded tail bound.

These results prove unobstructedness and show that the centre and cocenter form dual vector bundles near A . The centre morphism, however, need not be smooth: the type D algebras $D_{n,2}$ admit marked centre deformations that do not lift to algebra deformations. Finally, for geometric deformations of the completed simple flop over any complete Noetherian local complex base, the relative contraction quotient of the fixed tilting generator is finite free. Its centre and cocenter are finite free and commute with arbitrary base change. This proves the corresponding Hua–Toda flatness assertions in the completed simple-flop setting.

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1 Introduction

Let $f : X \rightarrow \operatorname{Spec} R$ be the completion at the contraction point of a simple flopping contraction of a smooth complex threefold, with exceptional curve $C \cong \mathbb{P}^1$. The contraction algebra of Donovan–Wemyss [6] is a finite-dimensional algebra that records noncommutative deformations of $\mathcal{O}_C(-1)$. Let $R \oplus N$ be the basic Van den Bergh generator, with N its nonfree summand. For $\Lambda = \operatorname{End}_R(R \oplus N)^{\operatorname{op}}$ and the free-summand projector e_0 ,

$$A = A_{\operatorname{con}} = \Lambda / \Lambda e_0 \Lambda.$$

We study the semisimple algebras to which A can deform.

Let $\ell \in \{1, \dots, 6\}$ be the generic length of the scheme-theoretic exceptional fibre. The contraction has genus-zero Gopakumar–Vafa invariants $n_1, \dots, n_\ell$, with $n_j = 0$ for $j > \ell$. These integers count the isolated $(-1, -1)$ -curves of class $j[C]$ in a generic local deformation [5, 26].

They determine the split semisimple algebra

$$S_{\text{GV}} = \prod_{j=1}^{\ell} M_j(\mathbb{C})^{\times n_j}. \quad (1.1)$$

Toda's noncommutative-width formula gives

$$\dim_{\mathbb{C}} A = \sum_{j=1}^{\ell} j^2 n_j = \dim_{\mathbb{C}} S_{\text{GV}} \quad [26, \text{Theorem 1.1}].$$

We ask whether a deformation realizes this numerical identity and whether every semisimple deformation of A has the same matrix blocks.

For finite-dimensional algebras A, B over an algebraically closed field k of characteristic zero, we say that A *deforms to* B if there is a finite free unital $k[[s]]$ -algebra with special fibre A and geometric generic fibre $B \otimes_k \overline{k((s))}$. Jelisiejew–Smoktunowicz prove that this formulation is equivalent to their split polynomial and orbit-closure formulations [16, Theorem 2.20].

Conjecture 1.1 (Wemyss; cf. [16, Questions 1.1–1.2]). *The contraction algebra A deforms to S_{GV} , and every semisimple complex deformation target of A is isomorphic to S_{GV} .*

Hua–Toda describe the matrix factors arising in a generic geometric deformation [14, Theorem 4.2] and formulate a flatness conjecture for contraction algebras and their degree-zero Hochschild (co)homology in geometric families [14, Conjecture 4.3]. Their noncommutative Donaldson–Thomas product formula recovers all GV invariants from the contraction algebra [14, Theorem 4.4 and Corollary 4.6]. The semisimple-deformation question asks for these invariants to appear as the multiplicities of matrix blocks.

Explicit algebraic constructions were developed by Bowman–Puljić–Smoktunowicz [2, Section 4], who construct a deformation of a nine-dimensional contraction algebra to $M_2(\mathbb{C}) \times \mathbb{C}^5$. Smoktunowicz studies strongly flat deformations and their semisimple specializations in [24]. Using the local normal forms of Brown–Wemyss [4], Jelisiejew–Smoktunowicz prove existence and uniqueness of the semisimple target for Types A and D [16, Theorem 1.3].

We prove existence and uniqueness uniformly for simple curve flops of smooth complex threefolds. The cocenter calculation also proves smoothness of the corresponding algebra law. We then study how centres vary in algebraic families and prove flatness of the relative contraction quotient in geometric families over arbitrary complete Noetherian local complex bases.

Main results

Theorem 1.2 (Semisimple existence). *Let $f : X \rightarrow \text{Spec } R$ be a completed simple flopping contraction of a smooth complex threefold, with contraction algebra A and GV invariants n_j . There is a finite free $\mathbb{C}[[s]]$ -algebra $\mathcal{A}$ such that*

$$\mathcal{A}/(s) \cong A, \quad \mathcal{A} \otimes_{\mathbb{C}[[s]]} K' \cong \prod_{j=1}^{\ell} M_j(K')^{\times n_j}$$

for a finite extension $K'/\mathbb{C}((s))$. After choosing a uniformizer, one may write $K' = \mathbb{C}((u))$ and $s = u^e$.

Section 2 constructs this family by pulling the universal marked tilting generator back along a morsification of the ADE classifying arc. Each coefficient- j root contributes a block $M_j(K')$, and there are n_j such roots. Finiteness and Toda's dimension formula then give freeness over the parameter DVR.

For a d -dimensional unital algebra, let X_d denote the affine scheme of associative multiplications on a fixed vector space with fixed identity, and put $G = \text{Stab}_{\text{GL}_d}(1)$.

Theorem 1.3 (Cocenter dimension and semisimple uniqueness). *In the setting of Theorem 1.2, let $d = \dim_{\mathbb{C}} A$. Then*

$$\dim_{\mathbb{C}} HH_0(A) = \dim_{\mathbb{C}} Z(A) = \sum_{j=1}^{\ell} n_j.$$

The point $A \in X_d$ is smooth, and $\overline{G \cdot S_{\text{GV}}}$ is the unique irreducible component through A . Every finite free formal deformation of A with geometrically semisimple generic fibre has geometric Wedderburn type S_{GV} .

Here $HH_0(A) = A/[A, A]$ is the Hochschild cocenter. Section 3 begins with the orbit criterion and then reduces uniqueness to the displayed cocenter dimension. The key calculation uses a square presentation of the universal marked cocenter: its determinant is the product of the reduced curve-divisor equations. Restriction to the classifying arc converts the determinant into the sum of the GV invariants. Symmetry, bimodule periodicity, and Tate duality then identify the tangent dimension at A with the dimension of the semisimple orbit. The resulting smoothness proves [16, Conjecture 1.4(1)] in the smooth simple-flop setting.

Unobstructedness and the centre

Jelisiejew–Smoktunowicz also conjectured that the centre of a contraction algebra varies well: its dimension should be locally constant near A , and the map sending an algebra to its centre should be smooth [16, Conjecture 1.4(2)]; see Conjecture 4.1. Section 4 makes this map precise as a morphism cent from a frame bundle over a neighbourhood of A in X_d to the scheme Com_c of commutative algebra structures with fixed identity. On tangent spaces it is governed by a linear map

$$\chi_B: HH^2(B) \longrightarrow T_{\text{com}}^1(Z(B)),$$

the centre tangent map of an algebra B near A , where T_{com}^1 is the Harrison group of first-order commutative deformations [13].

Theorem 1.4 (Unobstructedness and the centre). *In the setting of Theorem 1.3, put $c = \sum_j n_j$.*

- (i) $\widehat{\mathcal{O}}_{X_d, A} \cong \mathbb{C}[[t_1, \dots, t_{d(d-1)-d+c}]]$; in particular, A is unobstructed.
- (ii) On a neighbourhood U of A in X_d , the centre and the cocenter of the tautological algebra are vector bundles of rank c whose formation commutes with arbitrary base change. A relative symmetrizing trace makes them dual to each other.
- (iii) For $B \in U(\mathbb{C})$, the centre morphism is smooth at B if and only if χ_B is surjective. This holds on a dense open subset of U containing the semisimple algebras.

- (iv) *For the contraction algebras $D_{n,2}$, $n \geq 2$, the map χ is not surjective. Their centres have a polynomial family of commutative deformations, inside the smoothable component, whose first-order part is not in the image of the centre tangent map. Hence the centre morphism is not smooth at $D_{n,2}$.*

The counterexample in (iv) follows from a symbolic argument valid for all $n \geq 2$. For $D_{2,2}$, exact linear algebra computes the full image of the centre tangent map and proves nonlifting even without a chosen identification of the special centre (Propositions 4.19 and 4.20). Part (ii) also reduces the Hochschild assertions in Hua–Toda’s flatness conjecture to flatness of the relative contraction algebra. We establish that remaining assertion next.

Flatness over arbitrary bases

The next theorem concerns the relative quotient of a given geometric family. It applies over nonreduced bases as well as over reduced ones.

Theorem 1.5 (Hua–Toda flatness for completed simple flops). *Let f_0 be a completed simple flop of a smooth complex threefold, with contraction algebra A_0 and GV invariants n_j . Let B be any complete Noetherian local complex algebra with residue field $\mathbb{C}$, and let f_B be a flat deformation of this contraction, completed along the central contraction. Form the relative contraction quotient $\mathcal{A}_B$ using a lift of the fixed basic tilting generator, preserving its structure-sheaf summand. Then*

$$\mathrm{rank}_B \mathcal{A}_B = \sum_j j^2 n_j, \quad \mathrm{rank}_B HH_0(\mathcal{A}_B/B) = \mathrm{rank}_B Z(\mathcal{A}_B) = \sum_j n_j,$$

and all three modules are finite free. The centre and cocenter commute with arbitrary scalar extension. The corresponding local statement holds over Noetherian algebraic or analytic bases when the relative quotient has finite support over the base.

The precise formal setting and proof are given in Theorem 5.5 and Corollary 5.7. The universal marked contraction quotient has a square free resolution whose determinant is $\prod_j P_j^{j^2}$. Its pullback along a family of classifying arcs is injective on the closed slice; a local flatness argument then controls nilpotent directions in B . Thus the result does not require a semisimple generic fibre or a prior GV-constancy theorem for arbitrary families. The disk case of algebra flatness was previously proved by Kawamata [19, Theorem 3.1(2)].

Geometry and exact computation

The simultaneous-resolution theory of Katz–Morrison and Bryan–Katz–Leung describes a simple flop by an arc in a marked ADE base [20, 5]. A small translation separates its intersections with the curve divisors into transverse roots. The resulting ordinary double points carry the local matrix factors in the existence proof (Figure 1). The degree of the extended determinant line bundle identifies the size of each factor.

Etingof–Rains construct a finite free central extension of the preprojective algebra [8]. We prove that the cocenter of its marked corner is free and compute the determinant of multiplication by the central element. The nodal calculations of Etingof–Latour–Rains provide the required closed-fibre dimensions for lengths 2, 3, 4, 6, and length one is immediate [9]. We establish the remaining coefficient-five E_8 case by an exact modular rank calculation together

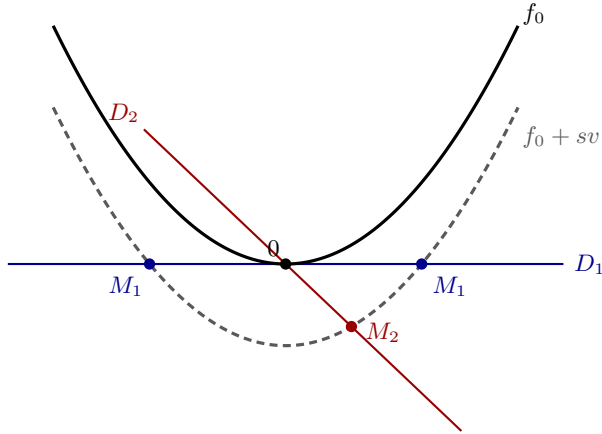


Figure 1: A schematic two-divisor slice of the marked base. A small translation separates a tangency with D_1 into transverse intersections; an intersection with D_j contributes the matrix block M_j . The counts for a given flop are the conserved intersection numbers of its classifying arc.

with a grading bound. Its relations, derived directly from the preprojective relations, and the numerical data needed for the proof are given in Section 3.6 and Appendix B.

All computer calculations in this thesis are exact. The programs are distributed with its source and described in Appendices A–C; the companion repository at commit [5a2001f](#) archives earlier E_8 material.

Conventions and organization. All algebras are associative and unital. The geometric arguments are over $\mathbb{C}$; Subsection 3.2 works over an algebraically closed characteristic-zero field. We write $A^e = A \otimes A^{\text{op}}$, $HH^i(A) = \text{Ext}_{A^e}^i(A, A)$, and $D = \text{Hom}_{\mathbb{C}}(-, \mathbb{C})$ in the complex setting. The notation $[A, A]$ always means the linear span of commutators; the commutative abelianization is the quotient by the ideal they generate. In the geometric construction, w is the surface-slice parameter and s is its external deformation parameter. In Section 3.5, z denotes the central element of the Etingof–Rains extension.

Section 2 constructs the semisimple deformation. Section 3 develops the orbit criterion, computes the marked cocenter, and proves uniqueness. Section 4 applies these results to the conjecture of Jelisiejew–Smoktunowicz on unobstructedness and on the centre. Section 5 proves the arbitrary-base geometric flatness theorem and conservation of the root schemes. The appendices give the root calculations, the E_8 relations and rank data, and the exact linear algebra for $D_{2,2}$.

2 Semisimple existence

Throughout this section the ground field is $\mathbb{C}$. Let

$$f: X \longrightarrow Y = \text{Spec } R$$

be the completion at the contraction point of a simple flopping contraction of a smooth complex threefold. Thus $\text{Exc}(f) = C \cong \mathbb{P}^1$, the scheme-theoretic length ℓ belongs to $\{1, \dots, 6\}$, and

the contraction algebra is denoted by A_{con} . We write n_j for the genus-zero Gopakumar–Vafa invariant in class $j[C]$.

Theorem 2.1 (Semisimple existence). *Let f be any completed simple one-curve flop of a smooth complex threefold, of length ℓ , with contraction algebra A_{con} and genus-zero GV invariants n_j . There exists a finite free $\mathbb{C}[[s]]$ -algebra $\mathcal{A}$ such that*

$$\mathcal{A}/(s) \cong A_{\text{con}}.$$

After a finite splitting extension $K'/\mathbb{C}((s))$,

$$\mathcal{A} \otimes_{\mathbb{C}[[s]]} K' \cong \prod_{j=1}^{\ell} M_j(K')^{\times n_j}. \quad (2.1)$$

We first construct a formal morsification with exactly n_j transverse roots of coefficient j . We then show that the universal marked tilting generator commutes with classifying base change and that each such root contributes the full matrix block M_j . Finally, we assemble the local blocks and prove finiteness over $\mathbb{C}[[s]]$. Toda’s dimension formula identifies the closed and generic fibre dimensions, which proves freeness over this DVR.

Definition 2.2 (Marked partial-resolution data). Let Φ be the primitive ADE root system of a general hyperplane section, with complex Cartan space $\mathfrak{h}$, Weyl group W , and simple roots $\Delta = \{\alpha_i\}$, determining the positive roots Φ^+ . The retained vertex is α_* . A marking fixes this vertex; on an endomorphism algebra, it fixes the corresponding summand projector. For a root $\alpha = \sum_i c_i(\alpha)\alpha_i$, its marked coefficient is $c_*(\alpha)$; the Levi roots are those with $c_*(\alpha) = 0$. Put

$$W_0 = \langle s_i : i \neq * \rangle, \quad H = \mathfrak{h}/W_0,$$

and let $\pi_W : H \rightarrow \mathfrak{h}/W$ be the induced finite map to the semiuniversal ADE base. Choose affine coordinates on H centred at the origin.

If θ is the highest root, the length classification gives

$$\ell = c_*(\theta)$$

[20]. The curve divisors and symbolic thickenings are indexed by $1 \leq j \leq \ell$ [5, Proposition 2.3 and Lemma 2.4]. Modulo the Levi root lattice, $[\alpha] = c_*(\alpha)[\alpha_*]$. The comparison with curve classes is expressed by degrees of extended line bundles in Lemma 2.8.

The six possibilities are shown in Figure 2.

Lemma 2.3 (The marked analytic classifying arc). *After replacing the source and target of the contraction by sufficiently small analytic neighbourhoods, there is a holomorphic map*

$$f_0 : \mathbb{D}_w \longrightarrow H, \quad f_0(0) = 0,$$

for which the given analytic contraction is isomorphic, as a marked contraction, to the pullback of the simultaneous partial resolution over H . Its completion is a formal classifying arc

$$\nu_0 : \text{Spf } \mathbb{C}[[w]] \longrightarrow H,$$

whose pullback is isomorphic to the original completed marked contraction.

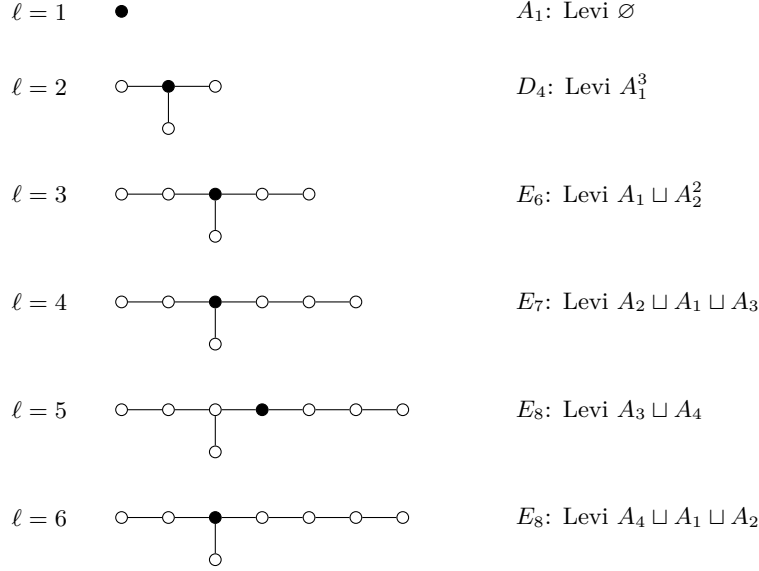


Figure 2: The six primitive marked ADE pairs. The filled vertex is the retained vertex α_* ; $\ell = c_*(\theta)$ is its coefficient in the highest root, and the Levi type is the diagram left after deleting it. For lengths 2, 3, 4, 6 the retained vertex is the branching vertex, which is the case treated by Etingof–Latour–Rains; length 5 is the one exception, and is the case settled by the computation of Proposition 3.14.

Proof. Choose a general analytic hyperplane parameter w through the contraction point. Its central fibre is the marked partial resolution of the ADE surface section. The semiuniversal deformation spaces of the full resolution, the marked partial resolution, and the ADE surface are, respectively,

$$\mathfrak{h}, \quad H = \mathfrak{h}/W_0, \quad \mathfrak{h}/W.$$

Bryan–Katz–Leung construct compatible analytic classifying maps from the w -disk and identify the resolved threefold neighbourhood with the Cartesian pullback of the simultaneous marked partial resolution [5, Section 2.1, especially Propositions 2.2–2.3]. Completing this Cartesian diagram at $w = 0$ gives the required isomorphism of marked formal contractions. $\square$

Let S_0 be the central partial-resolution surface, and write $\mathcal{I}_C \subset \mathcal{O}_{S_0}$ for the ideal of C in that surface.

Proposition 2.4 (The marked family and GV multiplicities). *For the classifying arc of Lemma 2.3, the simultaneous partial resolution has the following properties.*

- (i) *The base H is smooth affine space, the quotient $\pi_W: H \rightarrow \mathfrak{h}/W$ is finite, and the universal simultaneous partial resolution*

$$\pi_\ell: \mathcal{Y}_\ell \longrightarrow \mathcal{X}_\ell = \text{Spec } \mathcal{R}_\ell$$

is projective with source and target flat over H . Its fibres have dimension at most one and

$$R\pi_{\ell*} \mathcal{O}_{\mathcal{Y}_\ell} \cong \mathcal{O}_{\mathcal{X}_\ell}.$$

The universal flopping algebra is finite over its centre $\mathcal{R}_\ell$.

(ii) The curve discriminant is a union of precisely ℓ reduced irreducible divisors

$$D^{\text{curv}} = D_1 \cup \dots \cup D_\ell.$$

At a general point of D_j , the partial-resolution surface contains one smooth (-2) -curve in class $j[C]$. These curves form a projective flat family $\mathcal{C}_j \rightarrow D_j$, and its central fibre is the symbolic thickening $\mathcal{C}_j$ defined by $\mathcal{I}_C^{(j)}$.

(iii) A sufficiently small boundary-proper generic perturbation of f_0 meets D_j in k_j points, counted with multiplicity, where k_j is the length of the local Hilbert scheme at $\mathcal{C}_j$.

(iv) Toda's pure thickening is

$$\mathcal{O}_{C^{(j)}} = (\mathcal{O}_{S_0}/\mathcal{I}_C^j)/Q,$$

where Q is the maximal zero-dimensional subsheaf, and n_j is the length of its local Hilbert scheme. A generic geometric deformation has exactly n_j disjoint $(-1, -1)$ -curves in class $j[C]$.

(v) The contraction algebra satisfies

$$\dim_{\mathbb{C}} A_{\text{con}} = \sum_{j=1}^{\ell} j^2 n_j.$$

Proof. The description of the bases and the marked curve divisors is the Katz–Morrison and Bryan–Katz–Leung simultaneous-resolution construction [5, Propositions 2.2–2.3 and Lemma 2.4]. Nabijou–Wemyss give the restricted-root formulation for general flopping contractions in [22, Theorem 2.6]. Bryan–Katz–Leung [5, Lemma 2.8] identify the number of transverse intersections with D_j with the multiplicity k_j of the isolated Hilbert point $[C_j]$. Toda's definition and deformation interpretation identify the integers n_j , and his noncommutative-width theorem gives the dimension formula [26, Section 2.4 and Theorem 1.1]. The equality of the two central thickenings is proved in Lemma 2.11 below. The relative-dimension bound, the identity $R\pi_{\ell*}\mathcal{O} = \mathcal{O}$, and finiteness of the universal algebra over its centre follow from the universal-flop construction [17, Section 2.3, Assumption 3.1, and Section 3.3]. $\square$

2.1 Restricted-root geometry and marked morsification

Let $q : \mathfrak{h} \rightarrow H$ be the quotient map. For a root α , let $H_\alpha \subset \mathfrak{h}$ be its root hyperplane. The group W_0 preserves the marked coefficient and permutes the non-Levi roots of each coefficient. If $\mathcal{O}$ is such an orbit, the square-free product

$$p_{\mathcal{O}} = \prod_{\alpha \in \mathcal{O}} \alpha$$

is W_0 -invariant and descends to an equation on H . The roots of marked coefficient j form a single W_0 -orbit $\mathcal{O}_j$ (Lemma 2.5), so $p_{\mathcal{O}_j}$ descends to a polynomial $P_j \in \mathcal{O}(H)$, with $q^*P_j = p_{\mathcal{O}_j}$, cutting out the reduced irreducible divisor D_j of Proposition 2.4(ii). Put

$$D^{\text{curv}} = \bigcup_{j=1}^{\ell} D_j;$$

this is the curve discriminant, the locus where the marked partial resolution has a complete exceptional curve.

2.1.1 Root divisors and transverse local models

Lemma 2.5 (The coefficient strata are single W_0 -orbits). *For each of the six primitive marked ADE pairs and each $1 \leq j \leq \ell$, the set*

$$\mathcal{O}_j := \{\alpha \in \Phi^+ : c_*(\alpha) = j\}$$

is a single W_0 -orbit. Its cardinalities are:

length ℓ	marked pair	$ \mathcal{O}_1 , \dots, \mathcal{O}_\ell $
1	A_1	1
2	D_4	8, 1
3	E_6	18, 9, 2
4	E_7	24, 18, 8, 3
5	$E_8, c_*(\theta) = 5$	40, 30, 20, 10, 4
6	$E_8, c_*(\theta) = 6$	30, 30, 20, 15, 6, 5

In particular $\sum_j |\mathcal{O}_j|$ is the number of positive roots with $c_(\alpha) > 0$, written N_p in Section 3.5, and each D_j is irreducible.*

Proof. The group W_0 preserves c_* , so it acts on each $\mathcal{O}_j$. The finite reflection calculation in Appendix A enumerates the roots and the W_0 -orbits, giving the displayed cardinalities and transitivity. Since W_0 acts transitively on $\mathcal{O}_j$, the descended equation $p_{\mathcal{O}_j}$ is irreducible on H : a proper W_0 -stable factor would correspond to a proper W_0 -stable subset of $\mathcal{O}_j$. Hence D_j is irreducible, and by the first assertion of Lemma 2.6 it is reduced. $\square$

Lemma 2.6 (Reduced root divisors and the higher-root locus). *Let α be a non-Levi root. The quotient $\mathfrak{h} \rightarrow H = \mathfrak{h}/W_0$ is étale at a general point of H_α . Consequently the divisor descended from every W_0 -orbit of non-Levi root hyperplanes is generically reduced.*

There is a closed subset $\Sigma^{\text{bad}} \subset D^{\text{curv}}$ of codimension at least two in H , containing:

- (i) *the singular loci of all curve-discriminant components;*
- (ii) *intersections of distinct curve-discriminant components;*
- (iii) *the images of intersections $H_\alpha \cap H_\beta$, where α is non-Levi and $\beta \neq \pm\alpha$.*

We call the points of $D^{\text{curv}} \setminus \Sigma^{\text{bad}}$ good. At each Cartan lift of a good point of D_j , exactly one root pair $\{\pm\alpha\}$ vanishes.

Proof. Choose

$$x \in H_\alpha \setminus \bigcup_{\beta \neq \pm\alpha} H_\beta.$$

Its stabilizer in W is $\{1, s_\alpha\}$. Since α is not a Levi root, $s_\alpha \notin W_0$, and the W_0 -stabilizer of x is trivial. A finite quotient in characteristic zero is étale at a point with trivial stabilizer. The product of the distinct linear root equations therefore descends with generic multiplicity one.

For $\beta \neq \pm\alpha$, the hyperplanes H_α and H_β are distinct, so their intersection has codimension two. A finite morphism preserves the dimension of an irreducible closed subset. Thus all images in (iii) have codimension at least two in H . Distinct reduced curve divisors have no common

codimension-one component, and the singular locus of a reduced divisor in the smooth space H has ambient codimension at least two. Taking the finite union proves the claim. Outside this set, each Cartan lift lies on precisely one root hyperplane. For A_1 , there are no intersections of distinct root hyperplanes and the bad locus is empty. $\square$

Lemma 2.7 (Quadratic reflection coordinate). *Let $\bar{x} \in D_j \setminus \Sigma^{\text{bad}}$, and choose a Cartan lift $x \in H_\alpha$. There are completed local coordinates $(\zeta, z_2, \dots, z_r)$ at $\bar{x}$ and $(t, z_2, \dots, z_r)$ at its image in $\mathfrak{h}/W$ such that*

$$\widehat{\mathcal{O}}_{H, \bar{x}} = \mathbb{C}[[\zeta, z_2, \dots, z_r]], \quad \widehat{\mathcal{O}}_{\mathfrak{h}/W, \pi_W(\bar{x})} = \mathbb{C}[[t, z_2, \dots, z_r]],$$

and $\pi_W^* t = \zeta^2$, while $D_j = (\zeta = 0)$.

Proof. The W_0 -stabilizer is trivial, while the full Weyl stabilizer is the order-two group generated by s_α . This reflection acts by $\zeta \mapsto -\zeta$ on the normal line to H_α and trivially along the hyperplane. The invariant completed local ring is therefore $\mathbb{C}[[\zeta^2, z_2, \dots, z_r]]$. $\square$

Lemma 2.8 (Specialization of the marked degree). *Let $\mathcal{L}$ be a line bundle on the marked partial-resolution family, and let E_b be its exceptional curve over a good point $b \in D_j$. Then*

$$\deg_{E_b} \mathcal{L}_b = j \deg_C \mathcal{L}_0.$$

Proof. At the generic point of C , the normal central surface has a DVR as local ring. The symbolic j -thickening has generic length j , hence fundamental cycle $[C_j] = j[C]$. It follows that $\deg_{C_j} \mathcal{L}_0 = j \deg_C \mathcal{L}_0$. In the projective flat curve family $\mathcal{C}_j \rightarrow D_j$ of Proposition 2.4, both $\mathcal{O}_{\mathcal{C}_j}$ and the restriction of $\mathcal{L}$ are flat over the base. Their fibrewise Euler characteristics are locally constant. Their difference, which is the degree of $\mathcal{L}$ on a fibre, is therefore constant on the connected base germ. This gives the equality; compare [26, Section 2.4, equation (15)]. $\square$

Lemma 2.9 (The transverse threefold is an ordinary double point). *At a point of $D_j \setminus \Sigma^{\text{bad}}$, the transverse surface deformation is*

$$xy - z^2 = t.$$

After pullback to the marked partial-resolution base it is

$$xy - z^2 = \zeta^2,$$

and hence, after $\xi_+ = z + i\zeta$, $\xi_- = z - i\zeta$, it is the ordinary double point

$$xy - \xi_+ \xi_- = 0.$$

If a one-dimensional classifying curve meets D_j transversely, its partial resolution contains one $(-1, -1)$ -curve in numerical class $j[C]$.

Proof. By the choice of Σ^{bad} , only the root subsystem $\{\pm\alpha\} \cong A_1$ vanishes. The simultaneous-resolution description is therefore, transversely to the root hyperplane, the semiuniversal A_1 -deformation [5, Proposition 2.2]. For completeness, the A_1 singularity is $xy - z^2 = 0$, its one-dimensional T^1 is generated by the constant class, and hence a miniversal equation is

$$xy - z^2 = t.$$

Lemma 2.7 gives $t = \zeta^2$. Since $(z + i\zeta)(z - i\zeta) = z^2 + \zeta^2$, the pulled-back equation is exactly $xy - \xi_+ \xi_- = 0$. On a transverse classifying curve, ζ is a local parameter. Using it as the curve coordinate absorbs the higher terms in the original parametrization and gives the same completed ODP equation. The partial resolution is one of the two small resolutions of the ODP; its exceptional curve has normal bundle $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$. The degree comparison in Lemma 2.8 identifies its class as $j[C]$. $\square$

2.1.2 Perturbing the classifying arc

The local models above describe transverse intersections. We next choose a perturbation that produces only such intersections while keeping all roots inside a fixed disk.

Lemma 2.10 (The central arc is not contained in the curve discriminant). *The image of f_0 is contained in no D_j . After shrinking to a closed disk $\overline{\mathbb{D}}_\rho$,*

$$f_0^{-1}(D^{\text{curv}}) \cap \overline{\mathbb{D}}_\rho = \{0\}, \quad f_0(\partial \mathbb{D}_\rho) \cap D^{\text{curv}} = \emptyset.$$

If P_j is a reduced equation of D_j , then

$$0 < m_j := \text{ord}_{w=0} P_j(f_0(w)) < \infty.$$

Proof. If the arc were contained in D_j , the pullback of the projective flat family $\mathcal{C}_j \rightarrow D_j$ would be a two-dimensional closed subset of the exceptional locus of the threefold contraction, contradicting $\text{Exc}(f) = C$. Thus each reduced local equation P_j pulls back to a nonzero holomorphic function whose zero at the origin is isolated. There are only finitely many j , so a common shrinking gives the last assertions. $\square$

Lemma 2.11 (Symbolic and pure thickenings coincide). *The Bryan–Katz–Leung symbolic thickening C_j equals Toda’s pure one-dimensional thickening $C^{(j)}$. Consequently their Hilbert multiplicities satisfy $k_j = n_j$.*

Proof. Work in a stalk $\mathcal{S}$ of the normal surface S_0 and let $\mathfrak{p} \subset \mathcal{S}$ be the height-one prime of C . By definition,

$$\mathfrak{p}^{(j)} = \mathfrak{p}^j \mathcal{S}_{\mathfrak{p}} \cap \mathcal{S}.$$

This is $\mathfrak{p}$ -primary, and $\mathfrak{p}^{(j)}/\mathfrak{p}^j$ is supported at closed points, hence has finite length. Since $\mathcal{S}/\mathfrak{p}^{(j)}$ has only the associated prime $\mathfrak{p}$, it has no nonzero finite-length submodule. In the exact sequence

$$0 \longrightarrow \mathfrak{p}^{(j)}/\mathfrak{p}^j \longrightarrow \mathcal{S}/\mathfrak{p}^j \longrightarrow \mathcal{S}/\mathfrak{p}^{(j)} \longrightarrow 0,$$

the first term is therefore the maximal finite-length submodule of $\mathcal{S}/\mathfrak{p}^j$, precisely Toda’s Q . Thus $(\mathcal{O}/I^j)/Q = \mathcal{O}/I^{(j)}$, and the local Hilbert rings are identical. Proposition 2.4(iii)–(iv) gives $k_j = n_j$. $\square$

Lemma 2.12 (Parametric transversality with boundary). *Let M be a compact smooth manifold with boundary, let $\mathcal{U}$ and N be smooth manifolds, and let $S_1, \dots, S_m \subset N$ be smooth locally closed submanifolds. If a smooth map $\Xi: M \times \mathcal{U} \rightarrow N$ and its boundary restriction are transverse to every S_i , then the parameters for which both slice maps are simultaneously transverse to all S_i form a residual dense subset of $\mathcal{U}$. If $\dim_{\mathbb{R}} M = 2$, such a slice misses every stratum of real codimension at least four, and its boundary misses every stratum of real codimension at least two.*

Proof. For one stratum put $Z = \Xi^{-1}(S)$. Transversality makes Z a smooth incidence manifold, with boundary incidence Z_∂ . A parameter is a regular value of $Z \rightarrow \mathcal{U}$ exactly when its slice is transverse to S ; the analogous statement holds on the boundary. Sard's theorem, applied on a countable exhaustion by relatively compact coordinate pieces, shows that the common regular values are residual and dense. Intersect these sets for the finitely many strata. The dimension formula for transverse inverse images gives the final avoidance statements. $\square$

Lemma 2.13 (A boundary-controlled constant translate). *There is an arbitrarily small vector $v \in H \cong \mathbb{C}^{\dim H}$ such that $f_v(w) = f_0(w) + v$ is transverse to every smooth codimension-one curve-discriminant stratum, avoids Σ^{bad} , and has no curve-discriminant point on $\partial\mathbb{D}_\rho$. The vector can be chosen so that, for some $\epsilon > 0$,*

$$f_0(w) + sv \notin D^{\text{curv}} \quad \text{for } w \in \partial\mathbb{D}_\rho, \quad |s| \leq 1 + \epsilon.$$

Proof. Choose a finite Whitney stratification of the curve discriminant adapted to Σ^{bad} . The translation map

$$\overline{\mathbb{D}}_\rho \times H \longrightarrow H, \quad (w, \xi) \longmapsto f_0(w) + \xi$$

and its boundary restriction are submersions. Apply Lemma 2.12. A two-real-dimensional disk avoids the codimension-at-least-four bad strata, and its one-dimensional boundary avoids every curve-discriminant stratum. Since the compact boundary of the central disk is disjoint from the closed curve discriminant, reduced local equations have a positive minimum there. Uniform continuity supplies a ball of translations preserving boundary nonvanishing; choose the transverse parameter inside a slightly smaller ball. $\square$

2.1.3 Root counts and the formal generic fibre

The chosen translation preserves the total intersection multiplicity with each divisor. We record these roots in polynomials and then pass to the formal generic fibre.

Lemma 2.14 (Conservation of the root intersections). *Put $F(s, w) = f_0(w) + sv$ and $G_j(s, w) = P_j(F(s, w))$. On a connected s -disk U containing 0 and 1, the zero scheme of $G_j(s, \cdot)$ is finite flat of degree m_j and is represented by a monic polynomial*

$$W_j(s, T) \in \mathcal{O}(U)[T], \quad \deg_T W_j = m_j, \quad W_j(0, T) = T^{m_j}.$$

Proof. Boundary nonvanishing makes the argument-principle integral

$$\frac{1}{2\pi i} \int_{\partial\mathbb{D}_\rho} \frac{\partial_w G_j(s, w)}{G_j(s, w)} dw$$

constant in s , with value m_j . For $r \geq 1$, the analogous integral with an extra factor w^r is holomorphic in s and equals the r -th power sum of the zeros, counted with multiplicity. Newton's identities therefore give holomorphic elementary symmetric functions on the whole parameter disk. Let $W_j(s, T)$ be the monic polynomial having these coefficients.

For each s , its roots and their multiplicities are exactly the zeros of $G_j(s, \cdot)$ in the w -disk. Local Weierstrass preparation at each zero implies that G_j/W_j extends holomorphically across the zero locus and is a unit. Thus the two equations define the same analytic scheme. The quotient by the monic polynomial is locally free with basis $1, T, \dots, T^{m_j-1}$, and boundary nonvanishing makes its support proper over the parameter disk. At $s = 0$, all m_j zeros lie at the origin, so $W_j(0, T) = T^{m_j}$. $\square$

Lemma 2.15 (Passage from one good fibre to the formal generic fibre). *Put $B = \mathbb{C}[[s, w]]$ and $K = \mathbb{C}((s))$. The polynomials $W_j \in K[T]$ are separable and pairwise coprime. The classifying map on $\text{Spec } B[1/s]$ avoids Σ^{bad} , and every root is transverse to the smooth part of its unique divisor D_j .*

Proof. At $s = 1$, the discriminant of each W_j and every pairwise resultant are nonzero. They are therefore nonzero holomorphic functions of s . Their Taylor series may be divisible by s , but become units after passing to K . This proves separability and coprimality.

For an irreducible component $S \subset \Sigma^{\text{bad}}$, let $I_S = F^{-1}(S)$. Since $S \subset D^{\text{curv}}$, this is a closed subspace of the finite root incidence $\bigcup_j \{G_j = 0\}$ from Lemma 2.14. Its image in U is a proper analytic subset, because its fibre at $s = 1$ is empty. The completed direct image of $\mathcal{O}_{I_S}$ at $s = 0$ therefore has finite length over $\mathbb{C}[[s]]$ and vanishes after inverting s . There are finitely many such components. Finally, at a smooth point of D_j , separability is exactly the condition $\partial_w G_j \neq 0$, hence transversality. $\square$

Proposition 2.16 (The conserved multiplicities are the GV invariants). *For each j ,*

$$m_j = n_j. \quad (2.2)$$

Thus the formal constant-translation family has exactly n_j coefficient- j roots.

Proof. Write $\tilde{f}$ for the boundary-proper generic perturbation of f_0 used by Bryan–Katz–Leung. For that perturbation, conservation of the Cartier intersection number gives $m_j = \#\tilde{f}^{-1}(D_j) = k_j$. Lemma 2.11 gives $k_j = n_j$. Lemma 2.14 gives the same intersection number for the constant-translation family. $\square$

Proposition 2.17 (Marked formal morsification). *There is a finite extension K'/K with valuation ring $\mathbb{C}[[u]]$, where $K' = \mathbb{C}((u))$ and $s = u^e$ for some $e \geq 1$. The completed local models at the generic curve points are small resolutions of split ordinary double points. Moreover,*

$$W_j(u^e, w) = \prod_{a=1}^{n_j} (w - w_{j,a}(u)), \quad w_{j,a}(u) \in u\mathbb{C}[[u]],$$

with all roots pairwise distinct, including roots belonging to different coefficients. The pullback of the universal partial resolution has exactly n_j pairwise disjoint $(-1, -1)$ -curves in class $j[C]$, and no other complete curves.

Proof. First split the finitely many separable polynomials W_j . The resulting curve points are geometrically ordinary double points by Lemma 2.9. Enlarge the field by a finite extension splitting their quadratic tangent cones. The formal Morse lemma then gives the split ODP equations, and the small resolutions correspond to their two rulings. Denote the resulting field by K' . The normalization of $\mathbb{C}[[s]]$ in K' is a complete equicharacteristic DVR with residue field $\mathbb{C}$, hence is $\mathbb{C}[[u]]$; after a unit adjustment of u , its ramified parameter is $s = u^e$. Every root is integral because W_j is monic, and $W_j(0, T) = T^{n_j}$ forces its residue to be zero, so $w_{j,a}(u) \in u\mathbb{C}[[u]]$. Separability and coprimality give all the asserted distinctness. Lemma 2.9 supplies one $(-1, -1)$ -curve of class $j[C]$ at each root; different roots lie in different w -fibres and are disjoint.

To see that these are all the complete curves, observe that the completed target is affine. A complete curve in the source maps to a point and hence lies in the exceptional locus; the

regular function w is constant on it. The curve discriminant D^{curv} is the full locus of complete exceptional curves. Every generic point of the formal disk on that locus is a good point of one unique D_j , where the unique curve was already counted, and the disk avoids the codimension-two bad locus. Hence the list is exhaustive. $\square$

Remark 2.18. The generic fibre of the morsification is formed over $\mathbb{C}[[s, w]][1/s]$ and its finite extensions, as in Lemma 2.15, and not over $\mathbb{C}((u))[[w]]$. In the latter ring $w - w_{j,a}(u)$ is a unit whenever $w_{j,a} \neq 0$, so the moving roots would be lost.

2.2 Universal tilting bundles and the coefficient- j block

The morsification determines the number and classes of the exceptional curves. We now determine the matrix size contributed by each curve. The argument first controls base change of the universal generator, then computes its summands and contraction quotient on a split ordinary double point.

2.2.1 The universal generator and base change

Karmazyn's universal length- ℓ partial resolution carries the basic zero-perverse generator and its positive dual

$$\mathcal{T} = \mathcal{O} \oplus \mathcal{T}', \quad \mathcal{P} = \mathcal{T}^\vee = \mathcal{O} \oplus \mathcal{P}'.$$

Set

$$\mathcal{A}_\ell = \text{End}_{\mathcal{Y}_\ell}(\mathcal{T})^{\text{op}} \cong \text{End}_{\mathcal{Y}_\ell}(\mathcal{P}), \quad e_0 = [\mathcal{O}].$$

The universal source, target, centre, and algebra are flat over the marked base $H = \text{Spec } \mathbb{H}_\ell$ by the universal-flop construction [17, Sections 2.3 and 3.3]. The universal target is affine of finite type over H . The positive nonfree summand is the normalized maximal extension

$$0 \longrightarrow \mathcal{O}^{\oplus(\ell-1)} \longrightarrow \mathcal{P}' \longrightarrow \mathcal{L} \longrightarrow 0, \quad \deg_C \mathcal{L} = 1. \quad (2.3)$$

Thus $\text{rk } \mathcal{P}' = \ell$ and $\deg_C \det \mathcal{P}' = 1$. $\mathcal{P}$ belongs to the (-1) -perverse heart, while $\mathcal{T}$ belongs to the zero-perverse heart. Duality fixes the distinguished free projector and turns the endomorphism algebra into its opposite.

Lemma 2.19 (Tor-independence for arbitrary classifying pullback). *Let B be a Noetherian $\mathbb{H}_\ell$ -algebra, and form*

$$\begin{array}{ccc} \mathcal{Y}_B & \xrightarrow{g} & \mathcal{Y}_\ell \\ \pi_B \downarrow & & \downarrow \pi \\ \mathcal{X}_B & \xrightarrow{h} & \mathcal{X}_\ell. \end{array}$$

This square is Tor-independent. Consequently

$$R\pi_{B*} Lg^* E \simeq Lh^* R\pi_* E \quad (2.4)$$

for every perfect E on $\mathcal{Y}_\ell$, and $R\pi_{B} \mathcal{O}_{\mathcal{Y}_B} = \mathcal{O}_{\mathcal{X}_B}$.*

Proof. Locally write the rings as $C \rightarrow A \rightarrow Q$, where A and Q are flat over $C = \mathcal{O}_H$. Then

$$Q \otimes_A^L (A \otimes_C B) \simeq Q \otimes_C^L B \simeq Q \otimes_C B.$$

Thus the higher Tor groups vanish, and the standard derived base-change theorem gives (2.4). $\square$

Lemma 2.20 (Base change of the perverse generator). *The pullbacks $\mathcal{P}_B = g^*\mathcal{P}$ and $\mathcal{T}_B = g^*\mathcal{T}$ are projective generators of the (-1) - and zero-perverse hearts of π_B , respectively, and induce the corresponding equivalences on bounded coherent derived categories.*

Proof. Van den Bergh's category $\mathfrak{V}$ consists of globally generated bundles M satisfying $R^1\pi_*(M^\vee) = 0$. A finite evaluation surjection for $\mathcal{P}$ remains surjective after pullback. Since $\mathcal{P} \in \mathfrak{V}$, $R\pi_*\mathcal{T} = (\pi_*\mathcal{T})[0]$. By Lemma 2.19,

$$R\pi_{B*}\mathcal{T}_B \simeq Lh^*(\pi_*\mathcal{T}).$$

The left side lies in $D^{\geq 0}$, while the right side lies in $D^{\leq 0}$; hence both are concentrated in degree zero and $R^1\pi_{B*}\mathcal{T}_B = 0$. Relative ampleness of $\det \mathcal{P}$, global generation, and the direct $\mathcal{O}$ -summand also survive. Van den Bergh's intrinsic projective-generator criterion now applies [27, Propositions 3.2.5–3.2.7], and the derived equivalences follow from [27, Corollary 3.2.8]. $\square$

Proposition 2.21 (Ordinary endomorphism base change). *There is a canonical isomorphism of marked algebras*

$$\mathrm{End}_{\mathcal{Y}_B}(\mathcal{T}_B)^{\mathrm{op}} \cong \mathcal{A}_\ell \otimes_{\mathbb{H}_\ell} B. \quad (2.5)$$

All higher self-Ext groups vanish. The isomorphism preserves e_0 , and the bundle pullbacks preserve ranks and determinants. The algebra identification remains valid after central localization and Noetherian completion.

Proof. Apply derived base change to the vector bundle $\mathcal{E}nd(\mathcal{T})$. Tilting gives

$$R\pi_*\mathcal{E}nd(\mathcal{T}) = \pi_*\mathcal{E}nd(\mathcal{T})[0].$$

Its module of sections is the $\mathbb{H}_\ell$ -flat module underlying $\mathcal{A}_\ell^{\mathrm{op}}$. Hence its derived pullback is its ordinary pullback in degree zero. Taking sections proves (2.5) and the Ext vanishing. The distinguished free summand is pulled back, and rank and determinant commute with pullback. Localization and Noetherian completion are flat, so the same argument applies after either operation. $\square$

Write $\Lambda_B = \mathcal{A}_\ell \otimes_{\mathbb{H}_\ell} B$. Its quotient by $\Lambda_B e_0 \Lambda_B$ is the relative contraction quotient.

Proposition 2.22 (The special fibre is the basic NCCR). *Let $B = \mathbb{C}[[s, w]]$ carry a classifying map whose restriction modulo s is the central arc. Localize Λ_B at the central contraction point of its centre and complete there. The reduction $\widehat{\Lambda_B}/(s)$ is isomorphic to the basic NCCR of the original flop, preserving the free idempotent e_0 .*

Proof. The algebra is finite over its Noetherian centre, so completion commutes with reduction modulo s . Lemma 2.3 identifies the resulting completed geometry with the original flop.

On the completed one-curve flop, Van den Bergh's classification gives exactly two indecomposable positive perverse projectives: $\mathcal{O}$ and the maximal extension $\mathcal{M}$, with $\mathrm{rk} \mathcal{M} = \ell$ and $\deg_C \det \mathcal{M} = 1$ [27, Proposition 3.5.4 and Theorem 3.5.5]. Write the reduced nonfree universal summand as $\mathcal{O}^{\oplus a} \oplus \mathcal{M}^{\oplus b}$. Determinant degree gives $b = 1$, while rank gives $\ell = a + \ell$, hence $a = 0$. Thus the reduced positive generator is $\mathcal{O} \oplus \mathcal{M}$; dualizing gives the selected zero-perverse basic generator and fixes e_0 . $\square$

2.2.2 The generator at a transverse root

We now work at a geometric generic coefficient- j point after the finite extension of Proposition 2.17. Denote the point by λ , its completed small resolution by $\hat{Y}_\lambda \rightarrow \text{Spec } \hat{R}_\lambda$, and its exceptional curve by C_λ . The local model is split over K' .

Lemma 2.23 (The split ODP projectives over the splitting field). *Let F be any characteristic-zero field and put*

$$R_F = F[[a_1, a_2, b_1, b_2]]/(a_1 a_2 - b_1 b_2), \quad I = (a_1, b_1).$$

On the small resolution $\pi_Z: Z = \text{Proj}_{R_F} \bigoplus_{n \geq 0} I^n \rightarrow \text{Spec } R_F$, let $\mathcal{Q} = \mathcal{O}_Z(1)$. Then $\mathcal{O}_Z \oplus \mathcal{Q}$ is the basic positive perverse projective generator. Every positive perverse projective vector bundle is a direct sum of $\mathcal{O}_Z$ and $\mathcal{Q}$, and

$$\text{rk } \mathcal{Q} = 1, \quad \deg_C \mathcal{Q} = 1, \quad \pi_{Z*} \mathcal{Q} = I.$$

These assertions remain true after completion along the exceptional curve.

Proof. Write $\theta = b_1/a_1$ on the a_1 -chart and $\theta^{-1} = a_1/b_1$ on the b_1 -chart. After removing the Rees torsion and completing along the exceptional curve, the charts and their gluing are

$$\begin{aligned} \hat{U}_{a_1} &= \text{Spf } F[\theta][[a_1, b_2]], \quad b_1 = a_1 \theta, \quad a_2 = \theta b_2, \\ \hat{U}_{b_1} &= \text{Spf } F[\theta^{-1}][[b_1, a_2]], \quad a_1 = \theta^{-1} b_1, \quad b_2 = \theta^{-1} a_2. \end{aligned}$$

Thus Z is the split Atiyah small resolution. Its exceptional curve is given by $a_1 = b_2 = 0$ on $\hat{U}_{a_1}$ and $b_1 = a_2 = 0$ on $\hat{U}_{b_1}$, so $C \cong \mathbb{P}_F^1$. The transition of the tautological quotient is the transition of $\mathcal{O}_{\mathbb{P}^1}(1)$; consequently $\mathcal{Q}|_C = \mathcal{O}_C(1)$. The quotient $R_F/I \cong F[[a_2, b_2]]$ and the depth lemma show that I is maximal Cohen–Macaulay, hence reflexive. The pushforward of $\mathcal{Q}$ is reflexive by Hartogs extension across C in the smooth small resolution. It agrees with I away from the vertex, so $\pi_{Z*} \mathcal{Q} = I$.

Here $N_{C/Z} \cong \mathcal{O}_C(-1)^{\oplus 2}$. On the n -th infinitesimal graded piece of the formal neighbourhood of C , the restrictions of $\mathcal{O}_Z$ and $\mathcal{Q}^{-1}$ are respectively direct sums of

$$\text{Sym}^n N_{C/Z}^\vee \cong \mathcal{O}_C(n)^{\oplus (n+1)} \quad \text{and} \quad \mathcal{O}_C(n-1)^{\oplus (n+1)}.$$

Their first cohomology vanishes for every $n \geq 0$. Induction over the infinitesimal thickenings, followed by the theorem on formal functions, therefore gives

$$R^1 \pi_{Z*} \mathcal{O}_Z = R^1 \pi_{Z*} \mathcal{Q}^{-1} = 0.$$

The algebraic small resolution of the affine quadratic cone is the total space of $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$; the charts above describe its completion along the zero section. The bundle $\mathcal{Q}$ is relatively ample and globally generated. Together with the cohomology vanishings and the structure-sheaf summand, these properties verify Van den Bergh’s projective-generator criterion [27, Propositions 3.2.5–3.2.7], both over the complete local target and on the formal completion.

Its endomorphism algebra is finite over the complete local ring R_F and is semiperfect. The two summands have local endomorphism rings $\text{End}(\mathcal{O}_Z) = R_F = \text{End}(\mathcal{Q})$, and their degrees on C distinguish them. Their projectors therefore form a complete set of pairwise inequivalent primitive idempotents. Projectives over a semiperfect ring are direct sums of the corresponding indecomposable projectives. The tilting equivalence therefore proves the decomposition of every projective in the heart. The two summands and their idempotents are defined over F , and the same argument applies to the formal completion. $\square$

Proposition 2.24 (Decomposition of the local projective generator). *Let $\mathcal{Q}$ be the unique nonfree indecomposable positive perverse projective on the completed ODP small resolution, normalized by $\mathrm{rk} \mathcal{Q} = 1$ and $\deg \det \mathcal{Q}|_{C_\lambda} = 1$. Then*

$$\widehat{\mathcal{P}}'_\lambda \cong \mathcal{O}^{\oplus(\ell-j)} \oplus \mathcal{Q}^{\oplus j}. \quad (2.6)$$

After pushforward to the ODP centre,

$$N_\lambda \cong \widehat{R}_\lambda^{\oplus(\ell-j)} \oplus I_\lambda^{\oplus j},$$

where I_λ is the rank-one Atiyah MCM module.

Proof. Proposition 2.21 and Lemma 2.20 apply after field specialization and completion. The roots have been split over K' , so the completed local model is the split ODP of Lemma 2.23 with $F = K'$. That lemma, followed by Krull–Schmidt, gives $\widehat{\mathcal{P}}'_\lambda \cong \mathcal{O}^{\oplus a} \oplus \mathcal{Q}^{\oplus b}$. Rank gives $a + b = \ell$. Determinant commutes with pullback, and Lemma 2.8 gives

$$\deg_{C_\lambda} \det \widehat{\mathcal{P}}'_\lambda = j \deg_C \det \mathcal{P}' = j.$$

The same degree on the decomposition is b . Thus $b = j$ and $a = \ell - j$. Lemma 2.23 identifies the pushforward of $\mathcal{Q}$ with the Atiyah module I_λ . $\square$

2.2.3 The local contraction quotient

The preceding decomposition reduces the contraction quotient to stable endomorphisms of the Atiyah module. For a module over a local ring R , write $\underline{\mathrm{End}}_R$ for endomorphisms modulo maps factoring through finite free R -modules.

Lemma 2.25 (Stable endomorphisms of the Atiyah module). *Let*

$$R_{\mathrm{odp}} = F[[a_1, a_2, b_1, b_2]]/(a_1 a_2 - b_1 b_2), \quad I = (a_1, b_1),$$

where F has characteristic zero. Then

$$\underline{\mathrm{End}}_{R_{\mathrm{odp}}}(I) \cong F, \quad \underline{\mathrm{End}}_{R_{\mathrm{odp}}}(I^{\oplus j}) \cong M_j(F).$$

Proof. Write $R = R_{\mathrm{odp}}$. Since I is rank-one reflexive over the normal domain R , its endomorphisms are multiplications by fractions. At each height-one prime, I is free of rank one; hence $(I : I) \subseteq \bigcap_{\mathrm{ht} p=1} R_p = R$. Thus $\mathrm{End}_R(I) = R$.

We claim that

$$I^{-1} = R + R(b_2/a_1).$$

The displayed right-hand side is contained in I^{-1} , since $(b_2/a_1)a_1 = b_2$ and $(b_2/a_1)b_1 = a_2$. Conversely, if $qI \subset R$, put $r = qa_1 \in R$. Then $b_1 r \in a_1 R$. In $R/(a_1) = F[[a_2, b_1, b_2]]/(b_1 b_2)$, the annihilator of b_1 is (b_2) . Thus $r = a_1 r_0 + b_2 r_1$, and $q = r_0 + (b_2/a_1)r_1$, proving the claim. Endomorphisms factoring through a free module form the ideal

$$II^{-1} = (a_1, b_1) + (b_2, a_2) = (a_1, a_2, b_1, b_2) = \mathfrak{m}.$$

Therefore $\underline{\mathrm{End}}_R(I) = R/\mathfrak{m} = F$, and matrices give the second assertion. $\square$

Proposition 2.26 (Comparison with the MCM generator). *For the completed ODP small resolution, pushforward gives an isomorphism*

$$\mathrm{End}_{\widehat{Y}_\lambda}(\widehat{\mathcal{P}}_\lambda) \cong \mathrm{End}_{\widehat{R}_\lambda}(\widehat{R}_\lambda^{\oplus(\ell-j+1)} \oplus I_\lambda^{\oplus j}) \quad (2.7)$$

which preserves the projector onto the distinguished free summand.

Proof. The completed ODP is a normal complete local Gorenstein threefold, and the small resolution is smooth, projective, birational, and crepant, with fibres of dimension at most one and $R\widehat{\pi}_*\mathcal{O} = \widehat{R}$. The positive bundle is the perverse projective generator by Lemma 2.20; its two required $R^1\widehat{\pi}_*$ -vanishings follow from the defining $\mathfrak{V}$ -condition and global generation. Van den Bergh's comparison theorem therefore identifies the resolved endomorphism algebra with that of the pushed-forward MCM generator [27, Lemma 3.2.9 and Proposition 3.2.10]. Naturality sends the direct-summand projector to the distinguished free projector. $\square$

Theorem 2.27 (The completed coefficient- j factor). *At every transverse coefficient- j point used in the marked formal morsification, the completion of the relative contraction quotient is the matrix algebra $M_j(K')$. The maximal ideal of the completed centre acts trivially on this factor.*

Proof. After the arbitrary classifying pullback, central localization and Noetherian completion are flat. For a finite algebra Λ with idempotent e_0 ,

$$\widehat{\Lambda/\Lambda e_0 \Lambda} \cong \widehat{\Lambda}/\widehat{\Lambda} e_0 \widehat{\Lambda}.$$

By Proposition 2.26, this is the endomorphism algebra of $\widehat{R}^{\ell-j+1} \oplus I^j$ modulo the ideal generated by one distinguished free projector. Every other free projector factors through that copy, so the quotient is the stable endomorphism algebra of I^j . Lemma 2.25 gives $M_j(K')$. The centre acts through the residue field, proving the last assertion. $\square$

2.3 Finiteness, global support, and the existence theorem

We have identified each local matrix factor. It remains to show that they account for the entire generic fibre and that the relative algebra is free over the deformation parameter. Put $B = \mathbb{C}[[s, w]]$, and let $\nu : \mathrm{Spec} B \rightarrow H$ be the completion of the constant-translation family $F(s, w) = f_0(w) + sv$ from Lemmas 2.13 and 2.14. Set

$$R_B = \mathcal{R}_\ell \otimes_{\mathbb{H}_\ell} B.$$

Let $\mathfrak{n} \subset R_B$ be the maximal ideal of the central contraction point (it contains s), and define the central completion

$$S_\nu = \widehat{(R_B)_\mathfrak{n}}.$$

Let $\mathcal{T}_\nu$ be the completed pullback of the tilting bundle. The corresponding algebra and contraction quotient are

$$\Lambda_\nu = \mathcal{A}_\ell \otimes_{\mathcal{R}_\ell} S_\nu \cong \mathrm{End}(\mathcal{T}_\nu)^{\mathrm{op}}, \quad \mathcal{A} = \Lambda_\nu / \Lambda_\nu e_0 \Lambda_\nu. \quad (2.8)$$

Proposition 2.28 (Identification of the special fibre). *There is a marked algebra isomorphism*

$$\mathcal{A}/(s) \cong A_{\mathrm{con}}.$$

Proof. Proposition 2.22 gives $\Lambda_\nu/(s) \cong \Lambda$, the selected basic NCCR of the original flop, preserving e_0 . Right exactness gives

$$\mathcal{A}/(s) \cong \frac{\Lambda}{\Lambda e_0 \Lambda} = A_{\text{con}}.$$

□

Lemma 2.29 (Topological finiteness). *Let $R_0 = \mathbb{C}[[s]]$, let*

$$S = R_0[[x_1, \dots, x_m]]/J$$

be complete Noetherian, and let M be finite over S . If M/sM is finite-dimensional over $\mathbb{C}$, then M is finite over R_0 .

Proof. Choose lifts $m_1, \dots, m_d$ of a $\mathbb{C}$ -basis of M/sM . For $m \in M$, successive reduction modulo sM gives

$$m = \sum_{i=1}^d \left(\sum_{a=0}^{q-1} c_{ia} s^a \right) m_i + s^q m^{(q)}, \quad c_{ia} \in \mathbb{C}.$$

The series $c_i(s) = \sum_{a \geq 0} c_{ia} s^a$ belong to R_0 . For each q , their tails are divisible by s^q , so

$$m - \sum_i c_i(s) m_i \in s^q M.$$

The ring S is Noetherian local, s lies in its maximal ideal, and M is finite over S . Krull intersection gives $\bigcap_q s^q M = 0$. Hence the m_i generate M over R_0 . □

Proposition 2.30 (Finiteness of the relative contraction algebra). *The algebra $\mathcal{A}$ is a finite $\mathbb{C}[[s]]$ -module.*

Proof. The universal target is of finite type over H , so its pullback is of finite type over $B = \mathbb{C}[[s, w]]$. Completion at the central contraction point therefore gives a ring of the form

$$S_\nu \cong \mathbb{C}[[s, x_1, \dots, x_m]]/J$$

for finitely many generators. Proposition 2.4(i) states that $\mathcal{A}_\ell$ is finite over $\mathcal{R}_\ell$. Base change and completion therefore make Λ_ν finite over S_ν , and its idempotent quotient $\mathcal{A}$ remains finite over S_ν .

By Proposition 2.28, $\mathcal{A}/(s) \cong A_{\text{con}}$ is finite-dimensional. Lemma 2.29 now gives finiteness over $\mathbb{C}[[s]]$. □

Lemma 2.31 (Fullness away from the root support). *Over the geometric generic marked slice, the relative contraction quotient vanishes away from the finitely many transverse curve-root points. Consequently it is a finite algebra supported on exactly those closed points.*

Proof. By Proposition 2.4(ii), D^{curv} is the full locus where a complete curve occurs. Away from D^{curv} , a positive-dimensional fibre of the projective morphism would contain a complete curve, so the morphism is quasi-finite and hence finite. Since $R\pi_{\ell*} \mathcal{O}_{Y_\ell} = \mathcal{O}_{X_\ell}$, a finite base-changed morphism is an isomorphism. Proposition 2.17 meets D^{curv} only at the good transverse roots and supplies exactly the curves already counted. Away from those points, every summand of

the pushed-forward vector bundle is consequently locally free. After shrinking at a local point it is a finite free module, so every matrix unit of its endomorphism algebra factors through the distinguished rank-one free summand. Hence e_0 is full and the idempotent quotient vanishes. At every transverse point, Theorem 2.27 supplies a nonzero matrix block. By Proposition 2.30, the generic quotient is a finite-dimensional algebra over the generic point, finite over its Artinian centre. Its support is therefore a finite set of closed points, and by the above it is exactly the set of transverse curve-root points. $\square$

Lemma 2.32 (Assembly of the local factors). *Let S be a Noetherian K' -algebra and A a finite S -algebra supported at K' -rational maximal ideals $p_1, \dots, p_r$, so $S/p_i = K'$. Suppose*

$$\widehat{A_{p_i}} \cong M_{d_i}(K')$$

and the maximal ideal of $\widehat{S_{p_i}}$ acts trivially. Then

$$A \cong \prod_{i=1}^r M_{d_i}(K').$$

In particular its scheme-theoretic support is reduced.

Proof. Completion $S_{p_i} \rightarrow \widehat{S_{p_i}}$ is faithfully flat. Since $p_i \widehat{A_{p_i}} = 0$, faithful flatness gives $p_i A_{p_i} = 0$. Hence A_{p_i} already equals its completion and is $M_{d_i}(K')$. Put $J = \bigcap_i p_i$. The finite module JA localizes to zero away from the support and also at each p_i , where all other maximal ideals become units. Thus $JA = 0$. The Chinese remainder theorem gives $S/J \cong (K')^r$; its central idempotents split A , and the local factors are the stated matrix algebras. $\square$

Proposition 2.33 (The geometric generic fibre). *After the finite splitting extension $K'/\mathbb{C}((s))$ of Proposition 2.17,*

$$\mathcal{A} \otimes_{\mathbb{C}[[s]]} K' \cong \prod_{j=1}^{\ell} M_j(K')^{\times n_j}. \quad (2.9)$$

Proof. There are exactly n_j coefficient- j points by Propositions 2.16 and 2.17. In the normalized ramified model of that proposition, $K' = \mathbb{C}((u))$, $s = u^e$, and every root lies in $u\mathbb{C}[[u]]$. Thus the split points are K' -rational points of the formal w -disk, all specializing to its central maximal ideal. Lemma 2.31 gives the complete support, and Theorem 2.27 identifies every completed factor with $M_j(K')$. Apply Lemma 2.32 simultaneously to all support points. $\square$

Lemma 2.34 (Freeness from the fibre dimensions). *Let M be a finite module over the DVR $\mathbb{C}[[s]]$. If*

$$\dim_{\mathbb{C}} M/sM = \dim_{\mathbb{C}((s))} M[1/s] = r,$$

then M is free of rank r .

Proof. The structure theorem gives

$$M \cong \mathbb{C}[[s]]^{\oplus r} \oplus \bigoplus_{h=1}^m \mathbb{C}[[s]]/(s^{e_h}).$$

The generic dimension is the free rank, while reduction modulo s has dimension $r + m$. Equality forces $m = 0$. $\square$

Proof of Theorem 2.1. Take the relative contraction quotient defined in (2.8). Proposition 2.28 gives its special fibre, and Proposition 2.33 gives its geometric generic product. In particular

$$\dim_{\mathbb{C}((s))} \mathcal{A}[1/s] = \sum_{j=1}^{\ell} j^2 n_j.$$

Proposition 2.30 gives finiteness over $\mathbb{C}[[s]]$. On the other hand, Proposition 2.4(v) and the special-fibre identification give

$$\dim_{\mathbb{C}} \mathcal{A}/(s) = \dim_{\mathbb{C}} A_{\text{con}} = \sum_{j=1}^{\ell} j^2 n_j.$$

Lemma 2.34 now proves that $\mathcal{A}$ is finite free of this rank. $\square$

3 The cocenter and semisimple uniqueness

The semisimple deformation constructed in Section 2 places the contraction algebra in the closure of a semisimple orbit. We prove uniqueness by comparing the dimension of that orbit with the tangent dimension at the contraction algebra. The comparison reduces to a cocenter calculation on the marked ADE base.

3.1 Statement and proof strategy

For the completed simple one-curve flop $f: X \rightarrow \text{Spec } R$, let $A = A_{\text{con}}$ be its contraction algebra with $d = \dim_{\mathbb{C}} A$, and let n_j be its genus-zero GV invariants. Put

$$S_{\text{GV}} := \prod_{j=1}^{\ell} M_j(\mathbb{C})^{\times n_j}.$$

Theorem 3.1 (General semisimple–GV uniqueness). *For every completed simple one-curve flop of a smooth complex threefold,*

$$\dim_{\mathbb{C}} HH_0(A) = \dim_{\mathbb{C}} Z(A) = \sum_{j=1}^{\ell} n_j. \quad (3.1)$$

Consequently A is a smooth point of the fixed-unit associative-law scheme X_d , the orbit closure $\overline{G \cdot S_{\text{GV}}}$ is the unique irreducible component of X_d through A , and S_{GV} is the unique split semisimple complex deformation target of A . More generally, if $\mathcal{D}$ is a finite free unital $\mathbb{C}[[u]]$ -algebra with $\mathcal{D}/u\mathcal{D} \cong A$ and geometrically semisimple generic fibre, then that generic fibre has geometric Wedderburn type S_{GV} .

We first establish the orbit criterion, then use symmetry and periodicity to express its numerical hypothesis in terms of $HH_0(A)$. The remaining calculation proceeds on the Cartan cover: we construct the universal marked cocenter, descend its presentation to the marked base, and restrict it to the original classifying arc. The determinant records the intersections with the reduced curve divisors.

3.2 The orbit criterion

We give a fixed-unit formulation of the orbit-dimension criterion used in [16, Section 1 and Theorem 2.14]. Let k be algebraically closed of characteristic zero, let $V = k \cdot 1 \oplus W$ have dimension d , and let X_d be the affine scheme of associative multiplications on V for which the distinguished vector 1 is the identity. The scheme X_d is defined by the associativity equations. The group preserving the identity is

$$G = \text{Stab}_{\text{GL}(V)}(1).$$

For a unital algebra Λ , the normalized Hochschild cochains are

$$C_{\text{norm}}^n(\Lambda, \Lambda) = \text{Hom}_k((\Lambda/k \cdot 1)^{\otimes n}, \Lambda).$$

We write Z_{norm}^n and B_{norm}^n for the cocycles and coboundaries of this complex.

Lemma 3.2 (Fixed-unit tangent spaces). *For $\Lambda \in X_d(k)$,*

$$T_\Lambda X_d = Z_{\text{norm}}^2(\Lambda, \Lambda), \quad T_\Lambda(G \cdot \Lambda) = B_{\text{norm}}^2(\Lambda, \Lambda),$$

and

$$\text{Lie Stab}_G(\Lambda) = \text{Der}(\Lambda).$$

Moreover, G is smooth, connected, and irreducible of dimension $d(d-1)$, and its orbits are precisely the unital isomorphism classes.

Proof. A dual-number multiplication $m + \varepsilon\phi$ retains the fixed unit exactly when ϕ is normalized; its linearized associativity equation is the normalized Hochschild cocycle equation. Differentiating transport by $1 + \varepsilon f$, with $f(1) = 0$, gives the normalized coboundaries, whose kernel consists of derivations.

Relative to $k \cdot 1 \oplus W$, every element of G has the unique form

$$\begin{pmatrix} 1 & \lambda \\ 0 & h \end{pmatrix}, \quad \lambda \in W^*, \quad h \in \text{GL}(W).$$

Thus $G \cong W^* \rtimes \text{GL}(W)$, proving the group assertions. A unital algebra isomorphism fixes the unique identity, so it belongs to G . $\square$

Lemma 3.3 (A family gives fixed-unit orbit closure). *Let $\mathfrak{o}$ be a complete equicharacteristic DVR with algebraically closed residue field k , a chosen coefficient field $k \subset \mathfrak{o}$, and fraction field K . Let $\mathcal{C}$ be a finite free unital $\mathfrak{o}$ -algebra with special fibre A . If S is a d -dimensional k -algebra and*

$$\mathcal{C}_K \otimes_K \overline{K} \cong S \otimes_k \overline{K},$$

then

$$A \in \overline{G \cdot S} \subset X_d.$$

Proof. A uniformizer identifies $\mathfrak{o}$ with $k[[\varpi]]$. The unit of $\mathcal{C}$ is unimodular: its reduction is the nonzero unit of A . Extend it to an $\mathfrak{o}$ -basis lifting the prescribed basis of A . The structure constants then define a morphism $\text{Spec } \mathfrak{o} \rightarrow X_d$ whose closed point is A .

The geometric generic-fibre isomorphism is specified by finitely many entries algebraic over K , and therefore descends to a finite extension K'/K . The integral closure $\mathfrak{o}'$ of $\mathfrak{o}$ in K' is a finite complete DVR. Its residue field is a finite extension of the algebraically closed field k , hence is k . The generic-fibre isomorphism fixes the unit, so the base-changed generic point is a $G(K')$ -translate of S .

Let $Y = \overline{G \cdot S} \subset X_d$ with its reduced closed structure. Every equation of Y pulls back to zero in K' , hence to zero in the domain $\mathfrak{o}'$. The base-changed arc therefore factors through Y , and its closed point is A . This proves the orbit-closure relation. $\square$

Proposition 3.4 (Openness of semisimple orbits). *If $S \in X_d(k)$ is split semisimple, then S is a smooth point of the scheme X_d and $G \cdot S$ is open in X_d .*

Proof. The algebra S is separable, hence projective as an S^e -module. Thus $HH^i(S, S) = 0$ for $i > 0$, and in particular

$$Z_{\text{norm}}^2(S, S) = B_{\text{norm}}^2(S, S).$$

By Lemma 3.2, the tangent space of X_d at S equals the orbit tangent space.

The orbit is a smooth locally closed subvariety. Indeed, in characteristic zero its stabilizer is smooth, and the orbit is the homogeneous space $G/\text{Stab}_G(S)$; cf. [3, Proposition 1.11]. Near S , write $\mathcal{O} = \mathcal{O}_{X_d, S}$ and the orbit local ring as $\mathcal{O}/I$. If the common tangent/orbit dimension is δ , then $\dim(\mathcal{O}/I) = \delta$, while the embedding dimension of $\mathcal{O}$ is δ . Hence $\dim \mathcal{O} = \delta = \text{edim } \mathcal{O}$, so $\mathcal{O}$ is regular and therefore a domain. Since $\mathcal{O}/I$ has the same dimension and I is prime, $I = 0$. Thus the orbit contains an open neighbourhood of S ; translating that neighbourhood by G proves that the orbit is open. $\square$

Theorem 3.5 (Orbit-theoretic smoothness and uniqueness). *Let $A, S \in X_d(k)$, with S split semisimple. Assume*

$$A \in \overline{G \cdot S}, \quad \dim T_A X_d = \dim(G \cdot S). \quad (3.2)$$

Then:

- (i) A is a smooth point of X_d ;
- (ii) $\overline{G \cdot S}$ is the unique irreducible component of X_d through A ;
- (iii) if S' is split semisimple and $A \in \overline{G \cdot S'}$, then $S' \cong S$.

Proof. By Proposition 3.4 and the irreducibility of G , the orbit $G \cdot S$ is irreducible and open. Its closure is therefore an irreducible component $\mathcal{Z}$ of dimension $\delta = \dim(G \cdot S)$. Since $A \in \mathcal{Z}$, $\dim_A X_d \geq \delta$. The second hypothesis of (3.2) gives

$$\dim_A X_d \leq \dim T_A X_d = \delta.$$

Thus the local dimension equals the embedding dimension, so $\mathcal{O}_{X_d, A}$ is regular. Over a characteristic-zero field this is equivalent to smoothness at the k -point.

A regular local ring is a domain and has one minimal prime. Hence exactly one irreducible component of X_d passes through A , namely $\mathcal{Z}$. If S' is another semisimple generalization, the same openness argument makes $\overline{G \cdot S'}$ an irreducible component through A , so it also equals $\mathcal{Z}$. The two orbits are nonempty open subsets of the irreducible space $\mathcal{Z}$, hence intersect. Group orbits are equal or disjoint, so $G \cdot S' = G \cdot S$. Lemma 3.2 identifies this equality with a unital algebra isomorphism. $\square$

Corollary 3.6 (Hochschild numerical criterion). *Let $A, S \in X_d(k)$, with S semisimple and $A \in \overline{G \cdot S}$. The dimension equality in Theorem 3.5 is equivalent to*

$$\dim \operatorname{Der}(A) - \dim HH^2(A) = d - \dim Z(S). \quad (3.3)$$

In particular, it holds if

$$\dim Z(A) = \dim Z(S), \quad \dim HH^1(A) = \dim HH^2(A).$$

Proof. Lemma 3.2 gives

$$\dim T_A X_d = d(d-1) - \dim \operatorname{Der}(A) + \dim HH^2(A).$$

For split semisimple S , all derivations are inner and $\dim \operatorname{Der}(S) = d - \dim Z(S)$. Therefore

$$\dim(G \cdot S) = d(d-1) - (d - \dim Z(S)),$$

where characteristic zero ensures that the stabilizer group scheme is smooth, so its dimension equals the dimension of its Lie algebra. This proves the equivalence. Finally, $\dim \operatorname{Der}(A) = d - \dim Z(A) + \dim HH^1(A)$. $\square$

3.3 Symmetry, periodicity, and Tate duality

We now return to the contraction algebra A over $\mathbb{C}$. The next four results will convert the Hochschild numerical criterion into a statement about its cocenter.

Proposition 3.7 (Symmetry). *The contraction algebra is symmetric: $A_{\text{con}} \cong DA_{\text{con}}$ as A_{con} -bimodules.*

Proof. The completed base R is an isolated cDV singularity. If N is the nonfree Van den Bergh summand, then $A = \underline{\operatorname{End}}_R(N)^{\text{op}}$. The stable endomorphism algebra is symmetric by [1, Theorem 2.6], and the opposite of a symmetric algebra is symmetric. $\square$

Proposition 3.8 (Bimodule four-periodicity). *In the stable category of A -bimodules,*

$$\Omega_{A^e}^4(A) \simeq A.$$

Proof. The base R is an isolated cDV singularity with a crepant resolution. For its contraction algebra, Jasso–Keller–Muro construct an exact sequence of A -bimodules

$$0 \longrightarrow A \longrightarrow P_3 \longrightarrow P_2 \longrightarrow P_1 \longrightarrow P_0 \longrightarrow A \longrightarrow 0$$

with projective middle terms [15, Theorem 3.2.5]; it represents their restricted universal Massey product in $\operatorname{Ext}_{A^e}^4(A, A)$. This gives the claim.

The periodicity can also be obtained from Dugas’s theorem. Let $\mathcal{B} = \operatorname{CM}(R)$, whose projective-injective objects are add R . Write the Van den Bergh tilting bundle on the smooth source X as $\mathcal{O}_X \oplus \mathcal{N}$, and put $N = f_* \mathcal{N}$. By [15, Theorem 1.3.1], N is a $2\mathbb{Z}$ -cluster-tilting object of $\underline{\operatorname{CM}}(R)$, with stable endomorphism algebra A^{op} . Thus $R \oplus N$ is a 2-cluster-tilting object of the Frobenius category $\mathcal{B}$. The hypersurface identity $\Omega_R^2 \simeq \operatorname{Id}$ gives the parameters $d_D = 2$ and $r_D = 1$ in [7, Theorem 1.1(3)]. If $A \neq \mathbb{C}$, the one-curve contraction algebra is local and nonsemisimple, so it has no semisimple block, and Dugas’s theorem gives a periodic projective bimodule resolution of A^{op} , hence of A , with period dividing $(d_D + 2)r_D = 4$. If $A = \mathbb{C}$, its stable bimodule class is zero and its positive Hochschild groups vanish. $\square$

Lemma 3.9 (Tate–Hochschild duality). *For every symmetric finite-dimensional algebra A over $\mathbb{C}$,*

$$(\widehat{HH}^{-n}(A))^\vee \cong \widehat{HH}^{n-1}(A) \quad (n \in \mathbb{Z}).$$

If A is four-periodic in the stable bimodule category, then

$$DHH^1(A) \cong HH^2(A), \quad \dim HH^1(A) = \dim HH^2(A). \quad (3.4)$$

Proof. For a symmetric algebra A , the enveloping algebra A^e is symmetric, and $\widehat{HH}^*(A) = \widehat{\text{Ext}}_{A^e}^*(A, A)$. The displayed duality is [21, equation (2.11)]. Taking $n = 3$ and using four-periodicity gives

$$DHH^1(A) \cong D\widehat{HH}^{-3}(A) \cong \widehat{HH}^2(A) = HH^2(A),$$

because positive Tate degrees agree with ordinary Hochschild cohomology. $\square$

Lemma 3.10 (Centre–cocenter duality). *If A is finite-dimensional symmetric with symmetrizing trace $\tau: A \rightarrow \mathbb{C}$, then*

$$Z(A) \xrightarrow{\sim} HH_0(A)^\vee, \quad z \mapsto (\bar{a} \mapsto \tau(za)).$$

Consequently,

$$\dim Z(A) = \dim HH_0(A). \quad (3.5)$$

Proof. For $z \in Z(A)$,

$$\tau(z[a, b]) = \tau(zab - zba) = \tau(zab - azb) = 0,$$

so the functional factors through $A/[A, A]$. Conversely, identify $A \cong DA$ by the symmetrizing form. If the functional associated to $c \in A$ annihilates all commutators, then for all a, b ,

$$0 = \tau(c[a, b]) = \tau((ca - ac)b).$$

Nondegeneracy forces $ca = ac$ for all a , so $c \in Z(A)$. This proves perfection. $\square$

3.4 Fixed-unit tangent and orbit dimensions

Apply the fixed-unit tangent calculation of Subsection 3.2 over $\mathbb{C}$. For $G = \text{Stab}_{\text{GL}_d}(1)$, we have $\dim G = d(d-1)$.

Lemma 3.11 (Tangent formula). *The Zariski tangent space is*

$$T_A X_d = Z_{\text{norm}}^2(A, A).$$

For a contraction algebra A ,

$$\dim T_A X_d = d(d-1) - \dim \text{Der}(A) + \dim HH^2(A) \quad (3.6)$$

$$= d(d-1) - d + \dim Z(A). \quad (3.7)$$

Proof. The tangent-space identification is Lemma 3.2. The dimension formula in the proof of Corollary 3.6 gives the first equality. Since

$$\dim \text{Der}(A) = d - \dim Z(A) + \dim HH^1(A),$$

the second follows from Lemma 3.9. $\square$

Lemma 3.12 (Semisimple orbit dimension). *For split semisimple $S_{\text{GV}} = \prod_j M_j(\mathbb{C})^{\times n_j}$,*

$$\dim Z(S_{\text{GV}}) = \sum_j n_j, \quad \dim \text{Der}(S_{\text{GV}}) = d - \sum_j n_j,$$

and

$$\dim(G \cdot S_{\text{GV}}) = d(d-1) - d + \sum_j n_j. \quad (3.8)$$

Proof. The centre of S_{GV} has one dimension for each simple block, and there are $\sum_j n_j$ blocks. All derivations of a semisimple algebra over $\mathbb{C}$ are inner, so $\dim \text{Der}(S_{\text{GV}}) = d - \dim Z(S_{\text{GV}})$. The orbit dimension is then the case $S = S_{\text{GV}}$ of the computation in the proof of Corollary 3.6, the stabilizer group scheme being smooth in characteristic zero. $\square$

Combining (3.7) and (3.8),

$$\dim T_A X_d - \dim(G \cdot S_{\text{GV}}) = \dim Z(A) - \sum_j n_j. \quad (3.9)$$

Thus Theorem 3.5 and (3.5) reduce Theorem 3.1 to

$$\dim HH_0(A) = \sum_j n_j.$$

3.5 Universal marked cocenter on the Cartan cover

We now construct the universal presentation needed for this cocenter calculation. Fix one of the six primitive marked ADE pairs (Φ, p) . The notation p denotes the retained vertex α_* of Section 2, so $m_p(\alpha) = c_*(\alpha)$. Let $\mathfrak{h}$ be the Cartan space of rank r , and put

$$W_0 = \langle s_i : i \neq p \rangle, \quad H = \mathfrak{h}/W_0.$$

Write

$$\tilde{R} = \hat{\mathcal{O}}_{\mathfrak{h},0} \cong \mathbb{C}[[\lambda_1, \dots, \lambda_r]].$$

For a positive root $\alpha = \sum_i m_i(\alpha) \alpha_i$, put

$$N_p := \#\{\alpha \in \Phi^+ : m_p(\alpha) > 0\}. \quad (3.10)$$

Choose an orientation Q of the finite Dynkin diagram, and let $\overline{Q}$ be its double. Over $\tilde{R}$, set

$$\Pi_\lambda = \tilde{R} \overline{Q} / \left(\sum_{a \in Q_1} [a, a^*] - \sum_i \lambda_i e_i \right).$$

We use the convention of [17, Proposition 4.15] and write $(\lambda, \alpha) = \sum_i m_i(\alpha) \lambda_i$. Let

$$\mathcal{D}_p = e_p \Pi_\lambda e_p.$$

Theorem 3.13 (The marked cocenter on the Cartan cover). *For every primitive marked ADE pair (Φ, p) , there is an exact sequence*

$$0 \longrightarrow \tilde{R}^{N_p} \xrightarrow{\Psi_p} \tilde{R}^{N_p} \longrightarrow HH_0(\mathcal{D}_p/\tilde{R}) \longrightarrow 0 \quad (3.11)$$

such that

$$\det \Psi_p = \eta \Delta_p, \quad \Delta_p := \prod_{\substack{\alpha \in \Phi^+ \\ m_p(\alpha) > 0}} (\lambda, \alpha), \quad \eta \in \tilde{R}^\times. \quad (3.12)$$

Proof. We construct Ψ_p as multiplication by the central element on the cocenter of the Etingof–Rains extension.

The central extension. Take $\mu = \rho$, normalized by $\rho_i = (\rho, \alpha_i) = 1$. In the notation of Etingof–Rains [8], the universal parameters x_i satisfy $\sum_a [a, a^*] = \sum_i x_i e_i$, and z is adjoined centrally. The coefficient ring for the regular weight ρ is

$$K_\rho = \mathbb{C}[x_1 - \rho_1 z, \dots, x_r - \rho_r z].$$

The fibre at a parameter λ is the algebra Π_λ^ρ with relations

$$\sum_a [a, a^*] = \sum_i (\rho_i z + \lambda_i) e_i.$$

Via the coordinates $\lambda_i := x_i - \rho_i z$, completing K_ρ at its augmentation gives the formal power-series ring $\mathbb{C}[[\lambda_1, \dots, \lambda_r]]$, which is the completed Cartan parameter ring $\tilde{R}$. Here the λ_i are base coordinates and z is a central element of the algebra. The central extension is finite free over K_ρ by [8, Theorem 3.2(i)].

Define the marked corner of the completed universal central extension by

$$E_p := e_p(\Pi[z] \hat{\otimes}_{K_\rho} \tilde{R}) e_p, \quad \mathbf{M}_p := HH_0(E_p/\tilde{R}).$$

Since the Etingof–Rains central extension is finite free over K_ρ , the corner E_p is a finite $\tilde{R}$ -module. Therefore $\mathbf{M}_p$, as a quotient of E_p , is also finite over $\tilde{R}$.

The closed fibre of E_p is $e_p \Pi_0^\rho e_p$.

Generic rank. Etingof–Rains prove generic semisimplicity in [8, Theorem 3.2(v) and the display immediately following it]. That display decomposes the central extension into the ordinary deformed preprojective factors indexed by positive roots; the cited Crawley–Boevey theorem used there gives the simple module of dimension vector α . Hence, after passing to a splitting field $L/\text{Frac}(\tilde{R})$,

$$(E_p)_L \cong \prod_{m_p(\alpha) > 0} M_{m_p(\alpha)}(L). \quad (3.13)$$

Each matrix algebra contributes one dimension to its cocenter, so

$$\dim_L(\mathbf{M}_p)_L = N_p. \quad (3.14)$$

The closed-fibre dimension. We need the special-fibre equality

$$\dim_{\mathbb{C}} HH_0(e_p \Pi_0^\rho e_p) = N_p. \quad (3.15)$$

We identify the cases supplied by the nodal-corner calculation of Etingof–Latour–Rains (ELR) [9].

At $\lambda = 0$, our normalization of ρ gives the central-extension relation used by ELR, who carry out their case analysis at the weight $\mu = \rho$ [9, Section 3.2]. We use the corner computations inside their proof, not only their statement about the whole algebra.

ELR call the branching vertex of a Dynkin diagram of type D or E the *nodal vertex*. Karmazyn’s six primitive flop presentations determine the retained vertex and the Levi Weyl subgroup W_0 . The retained vertices and Levi types follow from [17, Examples 4.22–4.27]. With that numbering, they are:

length	Dynkin type	retained vertex	W_0 in Karmazyn	deleted diagram	ELR status
1	A_1	unique	trivial	empty	immediate
2	D_4	$p = 4$	S_2^3	A_1^3	nodal D
3	E_6	$p = 6$	$S_2 \times S_2^2$	$A_1 \sqcup A_2^2$	nodal E_6
4	E_7	$p = 7$	$S_3 \times S_2 \times S_4$	$A_2 \sqcup A_1 \sqcup A_3$	nodal E_7
5	E_8	$p = 4$	$S_4 \times S_5$	$A_3 \sqcup A_4$	non-nodal
6	E_8	$p = 8$	$S_5 \times S_2 \times S_3$	$A_4 \sqcup A_1 \sqcup A_2$	nodal E_8

For a positive root α , ELR’s phrase “the root contains the nodal simple root” is equivalent to $m_p(\alpha) > 0$. Hence their root count is N_p .

For type D , ELR [9, Section 3, Lemma 3.4] prove the required *degreewise upper bounds* for the cocenter of the nodal corner. Summing those bounds gives

$$\dim_{\mathbb{C}} HH_0(e_p \Pi_0^\rho e_p) \leq N_p.$$

On the other hand, the generic decomposition (3.14) gives a finite $\tilde{R}$ -module M_p of generic rank N_p . For any finite module over the local domain $\tilde{R}$, the dimension of the closed fibre is at least the generic rank. Therefore the reverse inequality holds and equality follows.

For the nodal E_6, E_7, E_8 corners, ELR [9, Section 3, Case 2b] compute the same graded cocenter dimensions by exact Magma calculations and obtain equality with the positive-root count. These are the primitive flop lengths 3, 4, 6 in the table.

For A_1 , the closed corner $e_p \Pi_0^\rho e_p = \mathbb{C}[z]/(z)$ has one-dimensional cocenter. Section 3.6 computes dimension 104 for the coefficient-five E_8 vertex. This proves (3.15) in all six cases.

Freeness. Choose lifts of a basis of the special fibre (3.15). Since HH_0 is the cokernel of the commutator map, it commutes with arbitrary base change. In particular,

$$M_p \otimes_{\tilde{R}} \mathbb{C} \cong HH_0(e_p \Pi_0^\rho e_p).$$

By (3.15), the right side has dimension N_p . Nakayama gives a surjection

$$\tilde{R}^{N_p} \twoheadrightarrow M_p.$$

After tensoring with $\text{Frac}(\tilde{R})$, (3.14) shows that this map is an isomorphism. Its kernel is a submodule of a free module over the domain $\tilde{R}$, hence torsion-free. Since its generic fibre is zero, the kernel is zero. Therefore

$$\mathbf{M}_p \cong \tilde{R}^{N_p}. \quad (3.16)$$

Multiplication by the central element. Multiplication by z induces

$$\mu_z: \mathbf{M}_p \rightarrow \mathbf{M}_p.$$

Setting $z = 0$ recovers the ordinary deformed preprojective corner, and since z is central,

$$HH_0(\mathcal{D}_p/\tilde{R}) \cong \mathbf{M}_p/z\mathbf{M}_p = \text{coker } \mu_z. \quad (3.17)$$

On the root factor α , the decomposition displayed immediately after [8, Theorem 3.2(v)] gives

$$z|_{V_\alpha} = -\frac{(\lambda, \alpha)}{(\rho, \alpha)}.$$

Therefore the action of μ_z on the one-dimensional cocenter of that factor has the same eigenvalue, and over the fraction field

$$\det \mu_z = \prod_{m_p(\alpha) > 0} \left(-\frac{(\lambda, \alpha)}{(\rho, \alpha)} \right).$$

The denominators are nonzero complex constants, so

$$\det \mu_z = \eta \Delta_p, \quad \eta \in \tilde{R}^\times.$$

This determinant is nonzero; hence μ_z is injective over the domain $\tilde{R}$. Together with (3.16)–(3.17), this gives the required presentation and determinant. $\square$

3.6 The coefficient-five E_8 computation

Deleting the coefficient-five vertex of E_8 leaves $A_3 \times A_4$, so

$$N_p = |\Phi^+(E_8)| - |\Phi^+(A_3)| - |\Phi^+(A_4)| = 120 - 6 - 10 = 104. \quad (3.18)$$

Let $B_5 = e_p \Pi_0^p e_p$ be the closed marked corner. Appendix B.1 derives its relations directly from the preprojective relations on the ρ -line. It constructs elements $\beta, \gamma \in B_5$ which, together with z , generate B_5 (Lemma B.2), and for which

$$\begin{aligned} P &= (\beta - 4z)(\beta - 5z)(\beta - 6z)(\beta - 7z), \\ R_4 &= (\gamma + \beta^2)^2 + \beta\gamma\beta - 5z^2(\gamma + \beta^2) + 4z^4, \\ R_5 &= \gamma\beta\gamma + \gamma^2\beta + \gamma\beta^3 - 5z^2\gamma\beta \end{aligned}$$

vanish in B_5 (Lemma B.3). Give $\mathbb{C}[z]\langle\beta, \gamma\rangle$ and B_5 the half-path grading

$$\deg z = \deg \beta = 1, \quad \deg \gamma = 2.$$

The three relations are homogeneous, so there is a surjection of graded algebras

$$Q_\rho := \frac{\mathbb{C}[z]\langle\beta, \gamma\rangle}{(P, R_4, R_5)} \twoheadrightarrow B_5. \quad (3.19)$$

We do not need to know whether it is an isomorphism. It induces a surjection $HH_0(Q_\rho) \twoheadrightarrow HH_0(B_5)$, so upper bounds for the cocenter of Q_ρ are upper bounds for that of B_5 . Remark B.4 explains why the signs of the parameter terms differ from a direct substitution into the displayed universal relations of [17, Example 4.26].

Proposition 3.14 (Exact modular calculation). *Let $h_m = \dim_{\mathbb{C}} HH_0(B_5)_m$. In degrees $0, \dots, 23$, the following sequence gives upper bounds for h_m :*

$$(1, 2, 4, 6, 7, 7, 7, 6, 6, 6, 6, 5, 5, 4, 4, 4, 4, 3, 3, 2, 2, 2, 2, 1).$$

The sum of these bounds is 99.

Proof. In degree m , use cyclic-word columns $z^a[w]_{\text{cyc}}$ with $a + \text{wt}(w) = m$. The rows are cyclic classes of homogeneous right multiples of P, R_4, R_5 . These span the relations in the cocenter: a two-sided multiple uRv has the same cyclic class as Rvu , and central powers of z can be moved to the front. The cokernel in degree m is therefore $HH_0(Q_\rho)_m$.

The relation matrices have integer entries. Exact Gaussian elimination over $\mathbb{F}_{1000003}$ gives the ranks recorded in Appendix B.2; the computation over $\mathbb{F}_{1000033}$ gives the same ranks. A nonzero modular minor is also nonzero over $\mathbb{Q}$, so the modular ranks are lower bounds for the characteristic-zero ranks. Subtracting them from the column counts bounds $\dim HH_0(Q_\rho)_m$, and hence h_m by (3.19). $\square$

3.6.1 Tail bound and exact grading comparison

Write h for the Coxeter number. In the Etingof–Rains grading, arrows have degree 1 and z has degree 2. Their Hilbert-series formulas [8, Section 2.1, equation (1), and Theorem 3.2(iii)–(iv)] show that the ordinary preprojective algebra has top arrow degree $h - 2$, while the ρ -special central extension has top degree $2(h - 2)$. On the marked corner (z, β, γ) have ER degrees $(2, 2, 4)$. The finite computation uses exactly half this grading. For E_8 , $h = 30$, so the top ER degree is $2(h - 2) = 56$, which is computational half-degree 28. Setting $z = 0$ leaves the ordinary preprojective corner; because the E_8 graph is bipartite, its closed paths have even arrow length, and the top arrow-degree bound $h - 2 = 28$ makes it vanish in half-degree above 14. Since z is central,

$$HH_0(B_5)/(z) \cong HH_0(B_5/(z)).$$

Consequently multiplication by z is surjective

$$HH_0(B_5)_{m-1} \twoheadrightarrow HH_0(B_5)_m \quad (m > 14).$$

The finite calculation gives $\dim HH_0(B_5)_{23} \leq 1$. Hence every degree $24, \dots, 28$ has dimension at most one. Therefore

$$\dim HH_0(B_5) \leq 99 + 5 = 104.$$

The finite module M_p has generic rank $N_p = 104$ by (3.14), and its closed fibre is $HH_0(B_5)$. The closed-fibre dimension is at least the generic rank, so

$$\dim HH_0(B_5) \geq 104.$$

Therefore

$$\dim HH_0(B_5) = 104 = N_p. \tag{3.20}$$

This proves (3.15) in the remaining case. Since the bounds add up to the exact dimension, every bound of Proposition 3.14 is attained, and each degree $24, \dots, 28$ contributes exactly one. In particular, the surjection $HH_0(Q_\rho)_m \rightarrow HH_0(B_5)_m$ is an isomorphism for $m \leq 23$.

3.7 Descent to the marked base

Write

$$\mathbb{H}_\Gamma = \mathcal{O}(\mathfrak{h}), \quad \mathbb{H}_\ell = \mathbb{H}_\Gamma^{W_0}, \quad R_H = \widehat{\mathbb{H}}_\ell, \quad \tilde{R} = \widehat{\mathbb{H}}_\Gamma,$$

with completions at the graded origins. The extension $R_H \rightarrow \tilde{R}$ is finite and faithfully flat. Indeed, W_0 is a reflection group, so both rings are regular local of the same dimension. A system of parameters of R_H is a system of parameters, hence a regular sequence, in the finite Cohen–Macaulay ring $\tilde{R}$. Auslander–Buchsbaum implies that $\tilde{R}$ is free over R_H ; its rank is positive.

3.7.1 The marked algebra on the Cartan cover

Let $\mathcal{A}_\Gamma$ be Karmazyn’s universal extended-Dynkin algebra and put $e_{0p} = e_0 + e_p$. Write $\mathcal{T}_{\Gamma,0p}$ for the two-summand generator on the Cartan partial resolution associated to this corner [17, Proposition 4.21], and $\mathcal{T}_\ell$ for the generator of Section 2.2. Completions below are taken at the central graded origins.

Proposition 3.15 (Comparison of the marked generators). *On the completed Cartan partial resolution there is an isomorphism*

$$\theta : \widehat{\mathcal{T}}_\ell \widehat{\otimes}_{R_H} \tilde{R} \xrightarrow{\sim} \widehat{\mathcal{T}}_{\Gamma,0p} \quad (3.21)$$

preserving the structure-sheaf and retained-vertex summands. It induces an isomorphism

$$\widehat{\mathcal{A}}_\ell \widehat{\otimes}_{R_H} \tilde{R} \xrightarrow{\sim} \widehat{e_{0p} \mathcal{A}_\Gamma e_{0p}} \quad (3.22)$$

sending e_0 to e_0 and e_p to e_p .

Proof. The Cartesian comparison in [17, Proposition 4.21] places the two generators on the same completed partial-resolution family over $\tilde{R}$. This family is projective over its affine target, and source and target are flat over the complete local Noetherian base. On its central fibre, both generators restrict to

$$\mathcal{T}_0 = \mathcal{O} \oplus \mathcal{T}_{0,p},$$

where the summands are indexed by the affine and retained vertices.

Each summand of $\mathcal{T}_0$ has vanishing higher self-Ext. Over the truncations of $\tilde{R}$ by powers of its maximal ideal, the obstruction to lifting such a bundle lies in Ext^2 , and the obstruction to extending an isomorphism between two lifts lies in Ext^1 . Both vanish, and the compatible lifts determine a lift over the completed family, which is projective over an affine base [18, Lemma 3.2]. Hence there are isomorphisms between the two lifts of $\mathcal{O}$ and between the two lifts of $\mathcal{T}_{0,p}$; we fix one of each. Their direct sum is θ , and it preserves both projectors. These isomorphisms are not canonical, since the summands have automorphisms, but only their existence is used. Taking opposite endomorphism algebras and applying Proposition 2.21 gives (3.22). An equivalence of module categories alone would only give a Morita equivalence; the summand-wise comparison identifies the algebras together with their idempotents. $\square$

For an algebra Λ with idempotent e_0 , scalar extension commutes with the quotient $\Lambda/\Lambda e_0 \Lambda$, by right exactness. The corner identity

$$\frac{e_{0p} \mathcal{A}_\Gamma e_{0p}}{(e_{0p} \mathcal{A}_\Gamma e_{0p}) e_0 (e_{0p} \mathcal{A}_\Gamma e_{0p})} \cong e_p (\mathcal{A}_\Gamma / \mathcal{A}_\Gamma e_0 \mathcal{A}_\Gamma) e_p \quad (3.23)$$

then identifies the marked quotient upstairs. By [17, Proposition 4.15], $\mathcal{A}_\Gamma$ is the universal extended-Dynkin deformed preprojective algebra. Killing its affine vertex leaves the finite-Dynkin algebra Π_λ : the affine parameter relation eliminates t_0 , and the remaining parameters are $\lambda_i = t_i$. Thus

$$\left(\widehat{\mathcal{A}}_\ell / \widehat{\mathcal{A}}_\ell e_0 \widehat{\mathcal{A}}_\ell\right) \widehat{\otimes}_{R_H} \widetilde{R} \cong e_p \Pi_\lambda e_p = \mathcal{D}_p. \quad (3.24)$$

3.7.2 The cocenter on the marked base

Put

$$\mathcal{C}_H = \widehat{\mathcal{A}}_\ell / \widehat{\mathcal{A}}_\ell e_0 \widehat{\mathcal{A}}_\ell, \quad \mathbf{M}_H = HH_0(\mathcal{C}_H / R_H).$$

The completion is at the graded origin. Since $\widetilde{R}$ is finite free over R_H , tensoring a complete module with $\widetilde{R}$ gives a finite direct sum of copies of that module and is already complete. Consequently (3.24) gives

$$\mathcal{C}_H \otimes_{R_H} \widetilde{R} \cong \mathcal{D}_p. \quad (3.25)$$

Theorem 3.16 (The universal marked cocenter). *The algebra $\mathcal{C}_H$ is finite over R_H . Its cocenter has a free resolution*

$$0 \longrightarrow R_H^q \xrightarrow{\Theta} R_H^q \longrightarrow \mathbf{M}_H \longrightarrow 0 \quad (3.26)$$

with

$$\mathrm{Fitt}_0^{R_H}(\mathbf{M}_H) = (\det \Theta) = (P_1 \cdots P_\ell),$$

where $P_j = 0$ defines the reduced coefficient- j curve divisor.

Proof. The algebra $\mathcal{D}_p$ is finite over $\widetilde{R}$. Finiteness descends along the faithfully flat Cartan cover, so $\mathcal{C}_H$ and $\mathbf{M}_H$ are finite over R_H . The commutator-cokernel description gives

$$\mathbf{M}_H \otimes_{R_H} \widetilde{R} \cong HH_0(\mathcal{D}_p / \widetilde{R}). \quad (3.27)$$

Take a finite free surjection onto $\mathbf{M}_H$, with kernel K . After base change, Theorem 3.13 makes $K \otimes_{R_H} \widetilde{R}$ projective. Finite projectivity descends, and a finite projective module over the local ring R_H is free. Moreover $\mathbf{M}_H$ has generic rank zero, because its Cartan pullback has a square presentation with nonzero determinant. The two free ranks therefore agree, giving (3.26).

Fitting ideals commute with base change, so

$$\mathrm{Fitt}_0^{R_H}(\mathbf{M}_H) \widetilde{R} = (\Delta_p). \quad (3.28)$$

For $i \neq p$, the simple reflection s_i preserves m_p and sends every positive root other than α_i to a positive root. Thus W_0 permutes the factors of Δ_p without signs, and $\Delta_p \in \widetilde{R}^{W_0} = R_H$. Contracting the ideal in (3.28) along the faithfully flat extension gives

$$\mathrm{Fitt}_0^{R_H}(\mathbf{M}_H) = (\Delta_p). \quad (3.29)$$

Grouping the roots by marked coefficient and using Lemmas 2.5 and 2.6 gives

$$\Delta_p = \eta P_1 \cdots P_\ell, \quad \eta \in R_H^\times. \quad (3.30)$$

□

The theorem does not identify $\mathbf{M}_H$ with the structure sheaf of the curve discriminant, nor does it claim that $\mathbf{M}_H$ is locally free on the discriminant where its components meet. Only the square presentation and its determinant are used below.

3.8 Restriction to the central arc

Let $R_1 = \mathbb{C}[[w]]$, and let $f_0 : \text{Spec } R_1 \rightarrow H$ be the classifying arc of the original flop. Set

$$\mathcal{C}_0 = \mathcal{C}_H \otimes_{R_H, f_0} R_1, \quad \mathbf{M}_0 = HH_0(\mathcal{C}_0/R_1).$$

By Lemma 2.10 and Proposition 2.16, the arc meets the curve discriminant only at the origin, and $\text{ord}_w P_j(f_0(w)) = n_j$.

3.8.1 The contraction algebra on the arc

Let $\widehat{\mathcal{R}}_\ell$ be the completed universal target ring at its graded origin, and put

$$Z_0 = \widehat{\mathcal{R}}_\ell \otimes_{R_H} R_1, \quad \Lambda_0 = \widehat{\mathcal{A}}_\ell \otimes_{R_H} R_1.$$

Then $\mathcal{C}_0 = \Lambda_0/\Lambda_0 e_0 \Lambda_0$. Let $\mathfrak{m}$ denote the contraction maximal ideal of Z_0 .

Lemma 3.17 (The central-arc quotient). *The natural completion map $\mathcal{C}_0 \rightarrow \widehat{\mathcal{C}}_{0\mathfrak{m}}$ is an isomorphism, and*

$$\mathcal{C}_0 \cong \widehat{\mathcal{C}}_{0\mathfrak{m}} \cong A. \quad (3.31)$$

In particular,

$$HH_0(\mathcal{C}_0) \cong HH_0(A). \quad (3.32)$$

Proof. The classifying arc is nonconstant: the power series $g(w) = P_1(f_0(w))$ has positive finite order. Weierstrass division makes $\mathbb{C}[[w]]$ finite over $\mathbb{C}[[g(w)]]$. The latter subring is contained in the image of R_H , so R_1 is finite over R_H .

Consequently Z_0 is finite over the complete Noetherian local ring $\widehat{\mathcal{R}}_\ell$. Modulo the maximal ideal of that ring its fibre is $R_1/\mathfrak{m}_H R_1$, where $\mathfrak{m}_H$ is the maximal ideal of R_H . This fibre is local Artinian with residue field $\mathbb{C}$. Thus Z_0 is local and complete at $\mathfrak{m}$. The finite Z_0 -algebra Λ_0 and its quotient $\mathcal{C}_0$ are also complete.

The ordinary tensor products above therefore compute the completed pullback along the central arc. Proposition 2.22 identifies Λ_0 with the original basic NCCR, preserving e_0 . Taking the idempotent quotient gives $\mathcal{C}_0 \cong A$, and then (3.32) follows by taking cocenters. $\square$

3.8.2 The length calculation

Base change gives $\mathbf{M}_0 = \mathbf{M}_H \otimes_{R_H} R_1$. Write $\Theta_0 = f_0^* \Theta$. Its determinant is

$$\det \Theta_0 = \eta(w) \prod_{j=1}^{\ell} P_j(f_0(w)) \neq 0. \quad (3.33)$$

Thus the base change of (3.26) is exact:

$$0 \longrightarrow R_1^q \xrightarrow{\Theta_0} R_1^q \longrightarrow \mathbf{M}_0 \longrightarrow 0. \quad (3.34)$$

Smith normal form over R_1 yields

$$\dim_{\mathbb{C}} \mathbf{M}_0 = \text{length}_{R_1} \mathbf{M}_0 = \text{ord}_w \det \Theta_0. \quad (3.35)$$

The relative and absolute cocenters have the same underlying $\mathbb{C}$ -vector space, since $r[a, b] = [ra, b]$ for central r . Combining Lemma 3.17, (3.33), and Proposition 2.16 gives

$$\dim_{\mathbb{C}} HH_0(A) = \sum_j \text{ord}_w P_j(f_0(w)) = \sum_j n_j. \quad (3.36)$$

Centre–cocenter duality then gives

$$\dim_{\mathbb{C}} Z(A) = \sum_j n_j. \quad (3.37)$$

3.9 Completion of the uniqueness proof

The cocenter calculation supplies the remaining hypothesis of the orbit criterion. We can now prove the theorem stated at the beginning of this section.

Proof of Theorem 3.1. Equations (3.9) and (3.37) give

$$\dim T_A X_d = \dim(G \cdot S_{\text{GV}}). \quad (3.38)$$

The general existence theorem of Section 2, combined with Lemma 3.3, supplies

$$A \in \overline{G \cdot S_{\text{GV}}}. \quad (3.39)$$

Theorem 3.5 now shows that A is smooth, that $\overline{G \cdot S_{\text{GV}}}$ is its unique component, and that any split semisimple algebra whose orbit closure contains A is isomorphic to S_{GV} .

For any finite free formal deformation $\mathcal{D}$ of A with geometrically semisimple generic fibre, let S' be the split $\mathbb{C}$ -algebra of that geometric Wedderburn type. Lemma 3.3 gives $A \in \overline{G \cdot S'}$, and the orbit criterion yields

$$S' \cong S_{\text{GV}} = \prod_{j=1}^{\ell} M_j(\mathbb{C})^{\times n_j}. \quad (3.40)$$

□

3.10 Relative cocenters

Corollary 3.18. *Let $\mathcal{B}$ be a finite free $\mathbb{C}[[s]]$ -algebra with special fibre A and geometrically semisimple generic fibre. Then $HH_0(\mathcal{B}/\mathbb{C}[[s]])$ is free of rank $\sum_j n_j$.*

Proof. The relative cocenter is finite and commutes with arbitrary base change. Theorem 3.1 gives dimension $\sum_j n_j$ on the closed fibre and, after a splitting extension, on the generic fibre. Lemma 2.34 proves freeness. □

Theorem 4.6 below removes the assumption on the generic fibre. For every finite free family of algebras over a Noetherian local $\mathbb{C}$ -algebra with residue field $\mathbb{C}$ and special fibre A , the relative centre and cocenter are free of rank $\sum_j n_j$, and their formation commutes with base change.

The cocenter along the morsification. For the geometric family used in the existence proof, the presentation gives a direct proof of this freeness statement. Pull back (3.26) along the morsification $F: \operatorname{Spec} B \rightarrow H$, where $B = \mathbb{C}[[s, w]]$, and write

$$M_F = \operatorname{coker}(\Theta_F : B^q \rightarrow B^q), \quad g = \det \Theta_F.$$

Its central determinant $g(0, w)$ is nonzero, of order $\sum_j n_j$. The adjugate identity gives $gM_F = 0$; Weierstrass preparation therefore makes M_F finite over $\mathbb{C}[[s]]$.

To check torsion-freeness, suppose $sv = \Theta_F y$ in B^q . Reducing modulo s and using the injectivity of Θ_0 gives $y = sy'$. Cancellation of s in the free module B^q then gives $v = \Theta_F y'$. Thus s is a non-zero-divisor on M_F , and M_F is free over $\mathbb{C}[[s]]$, of rank $\sum_j n_j$. This is the relative cocenter of the marked quotient obtained by scalar extension from $\mathcal{C}_H$.

Section 4 extends centre and cocenter freeness to all algebra deformations near A . Section 5 then uses the same presentation method to treat geometric deformations over arbitrary complete Noetherian local bases.

4 Unobstructedness and the centre morphism

The preceding section proves smoothness of the algebra law at a contraction algebra. We now study its consequences for deformations of the algebra and of its centre. The two questions appear together in the following conjecture of Jelisiejew–Smoktunowicz.

Conjecture 4.1 (Jelisiejew–Smoktunowicz [16, Conjecture 1.4]). *Let A be a contraction algebra and $d = \dim A$.*

- (1) *The algebra A defines a smooth point of the moduli space of d -dimensional unital associative algebras; in other words, A is unobstructed.*
- (2) *The dimension of the centre is locally constant near this point, and the map sending an algebra to its centre is smooth.*

Centres can jump in algebraic families, and the usual sufficient condition for unobstructedness does not apply here: $HH^3(A)$ is already nonzero for small contraction algebras. Jelisiejew–Smoktunowicz bound the local dimension from below by a semisimple deformation, reduce assertion (1) to the equality $\dim HH^1(A) = \dim HH^2(A)$, and verify it by computer for small d [16, Section 1]. This is the orbit-dimension argument of Subsection 3.2.

In this section we settle the conjecture for contraction algebras of simple flops of smooth threefolds, the setting of Sections 2–3. Assertion (1) holds (Theorem 4.4). The first half of assertion (2) holds in a stronger form: near A , the centre and the cocenter are vector bundles whose formation commutes with arbitrary base change (Theorem 4.6). The second half of assertion (2) fails. For the contraction algebras $D_{n,2}$ of [16], a first-order commutative deformation of the centre cannot lift to the algebra compatibly with the identification of the special centre (Theorem 4.16). Theorem 4.13 characterizes the locus where the centre morphism is smooth.

4.1 Moduli of algebras and the centre morphism

The moduli space in Conjecture 4.1 is the affine scheme of structure constants on a fixed d -dimensional space whose first basis vector is the identity [16, Section 2.1 and Proposition 2.8].

This is the scheme X_d of Subsection 3.2: it is defined by the associativity and unit equations, and it is not replaced by its reduction. Gerstenhaber identified the infinitesimal deformations of an algebra with its second Hochschild cohomology [11], and Gabriel studied the geometry of such schemes of algebra structures [10]; see [16, Section 1.1] for a survey. As before, $G = \text{Stab}_{\text{GL}_d}(1)$. We write $\mathcal{V} = V \otimes \mathcal{O}_{X_d}$ for the tautological algebra over X_d , and $\mathcal{V}_U$ for its restriction to an open subscheme $U \subset X_d$.

For $c \geq 1$, let Com_c be the affine scheme of commutative associative multiplications on $\mathbb{C}^c$ with identity e_1 , and let $G_c = \text{Stab}_{\text{GL}_c}(e_1)$, a smooth connected group of dimension $c(c-1)$. The scheme Com_c is the fixed-unit version of Poonen's moduli space of based commutative algebras [23]. For a commutative algebra Z , write $Z_{\text{com}}^2(Z)$ for the symmetric normalized Hochschild 2-cocycles of Z with values in Z , and $B_{\text{com}}^2(Z) = \delta(C_{\text{norm}}^1(Z, Z))$, where $(\delta f)(a, b) = af(b) - f(ab) + f(a)b$. Put

$$T_{\text{com}}^1(Z) = Z_{\text{com}}^2(Z) / B_{\text{com}}^2(Z).$$

This is the Harrison cohomology group that classifies first-order commutative deformations of Z [13].

Lemma 4.2 (Commutative tangent spaces). *Let $Z \in \text{Com}_c(\mathbb{C})$. Then*

$$T_Z \text{Com}_c = Z_{\text{com}}^2(Z), \quad T_Z(G_c \cdot Z) = B_{\text{com}}^2(Z),$$

and $T_{\text{com}}^1(Z)$ is a subspace of $HH^2(Z)$. The G_c -orbits are the isomorphism classes. The orbit $G_c \cdot \mathbb{C}^c$ of the split étale algebra is open in Com_c and has dimension $c(c-1)$.

Proof. The argument of Lemma 3.2 applies verbatim; a dual-number multiplication $m + \varepsilon\psi$ is commutative exactly when ψ is symmetric. Since Z is commutative, every normalized coboundary is symmetric, so $B_{\text{com}}^2(Z)$ is the space of all normalized coboundaries and $T_{\text{com}}^1(Z) \subset HH^2(Z)$. The algebra $\mathbb{C}^c$ is separable, so $HH^2(\mathbb{C}^c) = 0$ and its tangent space equals the orbit tangent space. Its stabilizer is the finite group permuting the primitive idempotents. The proof of Proposition 3.4 now shows that the orbit is open, of dimension $\dim G_c = c(c-1)$. $\square$

We call the closure of $G_c \cdot \mathbb{C}^c$, with its reduced structure, the *smoothable component* $\text{Sm}_c \subset \text{Com}_c$. It is irreducible of dimension $c(c-1)$.

Assertion (2) of Conjecture 4.1 combines two statements of different nature. Local constancy of $\dim Z$ concerns families of algebras that are already given: their centres should form a family of the right rank. Smoothness of the map to the centre is a lifting property in the opposite direction: every deformation of the centre should come from a deformation of the algebra. The first statement does not imply the second. The target of the map is not specified in [16]. The motivation given there is that deformations of the commutative algebra $Z(A)$ are better understood, so that unobstructedness would reduce the deformations of A to the commutative case [16, Section 1]. This points to all commutative deformations of the centre, and we take the target to be Com_c . Requiring the deformed centre to carry additional structure would change the question; Corollary 4.17 shows that smoothness still fails after restricting the target to the smoothable component or to the scheme-theoretic image.

Definition 4.3 (The centre morphism). Let $U \subset X_d$ be an open subscheme on which the centre $Z_U = Z(\mathcal{V}_U)$ is a locally free $\mathcal{O}_U$ -module of rank c whose formation commutes with arbitrary base change. Let $\mathcal{F}_U \rightarrow U$ be the scheme of unit-preserving frames: over a U -scheme

U' , its points are the isomorphisms $\mathcal{O}_{U'}^c \rightarrow Z_U \otimes \mathcal{O}_{U'}$ sending e_1 to 1. The unit section spans a subbundle of Z_U , so $\mathcal{F}_U$ is a G_c -torsor that is locally trivial for the Zariski topology. Transporting the multiplication of Z_U along the tautological frame defines a G_c -equivariant morphism

$$\text{cent}: \mathcal{F}_U \longrightarrow \text{Com}_c.$$

We say that the *centre morphism is smooth at $B \in U(\mathbb{C})$* if cent is smooth at a point of $\mathcal{F}_U$ over B . By equivariance, this does not depend on the choice of the point.

Equivalently, the morphism $U \rightarrow [\text{Com}_c/G_c]$ classifying Z_U is smooth at B : the frame bundle $\mathcal{F}_U$ is its base change along the smooth atlas $\text{Com}_c \rightarrow [\text{Com}_c/G_c]$.

4.2 Unobstructedness

Theorem 4.4 (Unobstructedness). *Let A be the contraction algebra of a completed simple flop of a smooth complex threefold, of any length, with GV invariants n_j . Put $d = \dim A$ and $c = \sum_j n_j$. Then*

$$\widehat{\mathcal{O}}_{X_d, A} \cong \mathbb{C}[[t_1, \dots, t_{d(d-1)-d+c}]]. \quad (4.1)$$

Consequently A is unobstructed. If $R' \rightarrow R$ is a surjection of local Artinian $\mathbb{C}$ -algebras with residue field $\mathbb{C}$, every finite free unital R -algebra $\mathcal{B}$ with $\mathcal{B} \otimes_R \mathbb{C} \cong A$ lifts to a finite free unital R' -algebra $\mathcal{B}'$ with $\mathcal{B}' \otimes_{R'} R \cong \mathcal{B}$. In particular, Conjecture 4.1(1) holds for these algebras.

Proof. By Theorem 3.1, A is a smooth point of X_d and $\overline{G \cdot S_{GV}}$ is the only component through it. Hence $\mathcal{O}_{X_d, A}$ is regular of dimension $\dim(G \cdot S_{GV}) = d(d-1) - d + c$ (Lemma 3.12). By Cohen's structure theorem, its completion, with residue field $\mathbb{C}$, is the power series ring (4.1).

For the lifting statement, choose an R -basis of $\mathcal{B}$ whose first vector is 1 and which lifts the fixed basis of A ; the unit is unimodular because it reduces to the unit of A . The structure constants define a homomorphism $\mathcal{O}(X_d) \rightarrow R$. It sends the maximal ideal of the point A into the nilpotent maximal ideal of R , so it factors through $\widehat{\mathcal{O}}_{X_d, A}$. Lift the images of the variables t_i arbitrarily to R' . The resulting homomorphism $\widehat{\mathcal{O}}_{X_d, A} \rightarrow R'$ gives structure constants over R' satisfying the associativity and unit equations, that is, an algebra $\mathcal{B}'$; it reduces to $\mathcal{B}$ by construction. $\square$

Remark 4.5. The proof does not use $HH^3(A)$; vanishing of $HH^3(A)$ would be a sufficient condition, but it fails in general [16, Section 1]. The number $d(d-1) - d + c$ is the dimension of the framed space of algebra laws with fixed identity. The deformation functor of A up to isomorphism has tangent space $HH^2(A)$, of dimension $\dim HH^1(A)$ by Lemma 3.9; the second assertion of Theorem 4.4 says that this functor is unobstructed.

4.3 The centre and the cocenter near a contraction algebra

Theorem 4.6 (Centre and cocenter near A). *Let A be as in Theorem 4.4, and $c = \sum_j n_j$. There is a smooth irreducible open neighbourhood U of A in X_d with the following properties.*

- (i) *The centre $Z_U = Z(\mathcal{V}_U)$ and the cocenter $M_U = HH_0(\mathcal{V}_U/\mathcal{O}_U)$ are locally free of rank c , and their formation commutes with arbitrary base change: for every morphism $g: U' \rightarrow U$, the natural maps $g^*Z_U \rightarrow Z(g^*\mathcal{V}_U)$ and $g^*M_U \rightarrow HH_0(g^*\mathcal{V}_U/\mathcal{O}_{U'})$ are isomorphisms.*

- (ii) Let τ_0 be a symmetrizing trace of A . After shrinking U , there is an $\mathcal{O}_U$ -linear map $\tau: \mathcal{V}_U \rightarrow \mathcal{O}_U$ that vanishes on commutators, restricts to τ_0 at A , and for which $(a, b) \mapsto \tau(ab)$ is a perfect pairing. Every fibre of $\mathcal{V}_U$ is then a symmetric algebra, and

$$Z_U \xrightarrow{\sim} \mathbf{M}_U^\vee, \quad z \mapsto (\bar{a} \mapsto \tau(za)).$$

In particular, the dimension of the centre is constant on U , as the first half of Conjecture 4.1(2) predicts.

Proof. By Theorem 3.1, A is a smooth point of X_d lying only on the component $\overline{G \cdot S_{\text{GV}}}$. Choose a connected open neighbourhood U of A contained in the smooth locus and meeting no other component. Then U is smooth, irreducible, and reduced, and $G \cdot S_{\text{GV}} \cap U$ is dense in U .

For the centre, consider the commutator operator

$$\kappa: \mathcal{V}_U \longrightarrow \mathcal{E}nd_{\mathcal{O}_U}(\mathcal{V}_U), \quad a \longmapsto (b \mapsto ab - ba),$$

a map of free modules of ranks d and d^2 with kernel Z_U . At a point B , its rank is $d - \dim Z(B)$. On the semisimple locus this is $d - c$, because S_{GV} has c simple factors, and at A it is $d - c$ by Theorem 3.1. All minors of size $d - c + 1$ vanish on the dense semisimple locus, hence on the reduced scheme U . Some minor of size $d - c$ is nonzero at A ; shrink U so that it is a unit. A matrix with an invertible $r \times r$ minor and vanishing $(r + 1)$ -minors is equivalent to $\begin{pmatrix} 1_r & 0 \\ 0 & 0 \end{pmatrix}$. Hence $\ker \kappa$ is a direct summand of rank c , and $\ker(g^* \kappa) = g^* \ker \kappa$ for every g . Since the centre of $g^* \mathcal{V}_U$ is $\ker(g^* \kappa)$, this proves the assertion for Z_U .

For the cocenter, apply the same argument to

$$b: \mathcal{V}_U \otimes_{\mathcal{O}_U} \mathcal{V}_U \longrightarrow \mathcal{V}_U, \quad a \otimes a' \longmapsto aa' - a'a,$$

whose cokernel is $\mathbf{M}_U$. Its rank is $d - \dim HH_0(B)$ at B , which equals $d - c$ on the semisimple locus and, by Theorem 3.1, at A . Thus $\mathbf{M}_U$ is locally free of rank c . Formation of a cokernel commutes with base change by right exactness.

For (ii), the trace τ_0 is an element of $HH_0(A)^\vee = \mathbf{M}_U^\vee \otimes \mathbb{C}(A)$, where we use that $\mathbf{M}_U$ is locally free. Extend it to a section of $\mathbf{M}_U^\vee$ on a neighbourhood of A , and compose with $\mathcal{V}_U \rightarrow \mathbf{M}_U$. This gives τ , which vanishes on commutators and restricts to τ_0 . The Gram determinant of $(a, b) \mapsto \tau(ab)$ in the fixed basis is a regular function that is nonzero at A ; shrink U so that it is a unit. Every fibre is then symmetric. The displayed map is well defined, because $\tau(z[a, b]) = 0$ for central z . By (i), its fibre at B is the map $Z(B) \rightarrow HH_0(B)^\vee$ of Lemma 3.10, an isomorphism. A map of locally free sheaves of the same rank that is an isomorphism on all fibres is an isomorphism. $\square$

Corollary 4.7 (Artinian deformations). *Let R be a local Artinian $\mathbb{C}$ -algebra with residue field $\mathbb{C}$, and let $\mathcal{B}$ be a finite free unital R -algebra with $\mathcal{B} \otimes_R \mathbb{C} \cong A$. Then $Z(\mathcal{B})$ and $HH_0(\mathcal{B}/R)$ are free R -modules of rank c , their formation commutes with every base change $R \rightarrow R''$ of local Artinian $\mathbb{C}$ -algebras, and every symmetrizing trace of A lifts to an R -linear symmetrizing trace of $\mathcal{B}$. In particular, every central element of A lifts to a central element of $\mathcal{B}$.*

Proof. A unit-preserving frame lifting the basis of A defines a morphism $\text{Spec } R \rightarrow X_d$ whose only point maps to A . It therefore factors through U , and Theorem 4.6 pulls back. $\square$

The corollary concerns all Artinian deformations of A , not only those along a chosen semisimple curve. It also gives a numerical identity for algebras near A that need not be contraction algebras.

Corollary 4.8. *For every $B \in U(\mathbb{C})$, $\dim HH^1(B) = \dim HH^2(B)$.*

Proof. The scheme U is smooth and irreducible of dimension $d(d-1) - d + c$, so $\dim T_B X_d = d(d-1) - d + c$. As in the proof of Corollary 3.6, $\dim T_B X_d = d(d-1) - \dim \operatorname{Der}(B) + \dim HH^2(B)$ and $\dim \operatorname{Der}(B) = d - \dim Z(B) + \dim HH^1(B)$. Since $\dim Z(B) = c$ by Theorem 4.6, the claim follows. $\square$

Remark 4.9 (Flatness of Hochschild invariants). Hua–Toda conjecture that, for every flat deformation of a flopping contraction over a base T , the contraction algebra and its zeroth Hochschild homology and cohomology are flat over T [14, Conjecture 4.3]. The zeroth Hochschild cohomology is the centre. For a generic geometric deformation, the generic cocenter has dimension $\sum_j n_j$ [14, Theorem 4.2]; Theorem 3.1 gives the same dimension at the central fibre. Once the relative contraction quotient is finite locally free, a local frame maps a neighbourhood of that fibre into U . Theorem 4.6 then gives local freeness and arbitrary base change for both Hochschild invariants. It remains to prove flatness of the relative quotient itself. Section 5 does this over complete Noetherian local bases and gives the corresponding local algebraic and analytic statement.

4.4 The centre tangent map and the smooth locus

Let U be as in Theorem 4.6, so that the centre morphism $\operatorname{cent}: \mathcal{F}_U \rightarrow \operatorname{Com}_c$ of Definition 4.3 is defined. Fix $B \in U(\mathbb{C})$ and put $Z_B = Z(B)$. For $\phi \in Z_{\operatorname{norm}}^2(B, B)$, let B_ϕ be $B[\varepsilon]$ with multiplication $aa' + \varepsilon\phi(a, a')$, where $\varepsilon^2 = 0$. By Lemma 3.2, these are the tangent vectors to X_d at B .

Lemma 4.10 (The centre tangent map). *Let $\phi \in Z_{\operatorname{norm}}^2(B, B)$.*

- (i) *There is a linear map $\tau: Z_B \rightarrow B$ with $\tau(1) = 0$ such that $z + \varepsilon\tau(z)$ is central in B_ϕ for all $z \in Z_B$; equivalently,*

$$[\tau(z), a] = \phi(a, z) - \phi(z, a) \quad (a \in B). \quad (4.2)$$

Any two such maps differ by a linear map $Z_B \rightarrow Z_B$ vanishing at 1.

- (ii) *For such τ , the bilinear map*

$$\psi(z, z') = \phi(z, z') + \tau(z)z' + z\tau(z') - \tau(zz') \quad (4.3)$$

takes values in Z_B and lies in $Z_{\operatorname{com}}^2(Z_B)$. The map $z + \varepsilon z'' \mapsto z + \varepsilon(\tau(z) + z'')$ is an isomorphism from $Z_B[\varepsilon]$, with multiplication $zz' + \varepsilon\psi(z, z')$, onto $Z(B_\phi)$.

- (iii) *The class of ψ in $T_{\operatorname{com}}^1(Z_B)$ depends only on the class of ϕ in $HH^2(B)$. This defines a linear map*

$$\chi_B: HH^2(B) \longrightarrow T_{\operatorname{com}}^1(Z_B),$$

the centre tangent map of B .

Proof. (i) The algebra B_ϕ corresponds to a morphism $\text{Spec } \mathbb{C}[\varepsilon] \rightarrow U$. By Theorem 4.6(i), $Z(B_\phi)$ is free of rank c over $\mathbb{C}[\varepsilon]$ and reduces to Z_B . Hence every z lifts to a central element $z + \varepsilon b$; choosing such lifts on a basis containing 1 defines τ . Comparing $(z + \varepsilon\tau(z))a$ with $a(z + \varepsilon\tau(z))$ in B_ϕ gives (4.2). An element εb is central in B_ϕ exactly when $b \in Z_B$, which proves the last assertion.

(ii) The product of the central elements $z + \varepsilon\tau(z)$ and $z' + \varepsilon\tau(z')$ is

$$zz' + \varepsilon(\phi(z, z') + \tau(z)z' + z\tau(z')) = (zz' + \varepsilon\tau(zz')) + \varepsilon\psi(z, z').$$

It is central, so $\varepsilon\psi(z, z')$ is central and $\psi(z, z') \in Z_B$. Central elements commute, so ψ is symmetric, and $\psi(1, z') = 0$ because ϕ is normalized and $\tau(1) = 0$. The displayed map is $\mathbb{C}[\varepsilon]$ -linear and bijective, since both sides are free of rank c and it reduces to the identity of Z_B . By the computation above it is multiplicative for the stated product. That product is therefore associative, so ψ is a Hochschild cocycle.

(iii) Changing τ by $f: Z_B \rightarrow Z_B$ with $f(1) = 0$ changes ψ by δf . If $\phi = \delta g$ with $g \in C_{\text{norm}}^1(B, B)$, then $1 + \varepsilon g$ is an algebra isomorphism $B_\phi \rightarrow B[\varepsilon]$. Hence $z - \varepsilon g(z)$ is central in B_ϕ , and the choice $\tau = -g|_{Z_B}$ in (4.3) gives $\psi = 0$. Both constructions are linear in ϕ . $\square$

Proposition 4.11 (The differential of the centre morphism). *Let $(B, \mathfrak{f})$ be a $\mathbb{C}$ -point of $\mathcal{F}_U$, and identify $\mathbb{C}^c$ with Z_B by the frame $\mathfrak{f}$. The image of $d\text{cent}: T_{(B, \mathfrak{f})}\mathcal{F}_U \rightarrow T_{Z_B}\text{Com}_c = Z_{\text{com}}^2(Z_B)$ is spanned by $B_{\text{com}}^2(Z_B)$ and the cocycles (4.3). Consequently*

$$\text{rank } d\text{cent}_{(B, \mathfrak{f})} = \dim B_{\text{com}}^2(Z_B) + \text{rank } \chi_B, \quad (4.4)$$

and $d\text{cent}_{(B, \mathfrak{f})}$ is surjective if and only if χ_B is surjective.

Proof. A tangent vector to $\mathcal{F}_U$ at $(B, \mathfrak{f})$ is a first-order deformation B_ϕ together with a unit-preserving frame of $Z(B_\phi)$ lifting $\mathfrak{f}$. By Lemma 4.10(i), these frames are $z \mapsto z + \varepsilon\tau(z)$, where τ runs through a torsor under $C_{\text{norm}}^1(Z_B, Z_B) = \text{Lie } G_c$. By Lemma 4.10(ii), $d\text{cent}$ sends this tangent vector to ψ . Varying τ gives all of $\psi + B_{\text{com}}^2(Z_B)$, which proves the description of the image. The image contains $B_{\text{com}}^2(Z_B)$, and its image in $T_{\text{com}}^1(Z_B)$ is the image of χ_B . $\square$

The rank formula includes the tangent directions coming from changes of frame. A direct comparison of $\dim Z_{\text{norm}}^2(B, B)$ with $\dim Z_{\text{com}}^2(Z_B)$ would not detect the relevant obstruction, since the two frame groups have different dimensions.

Lemma 4.12 (A tangent criterion for smoothness). *Let $g: X \rightarrow Y$ be a morphism of schemes of finite type over $\mathbb{C}$, let $x \in X(\mathbb{C})$ and $y = g(x)$, and suppose that X is smooth at x . Then g is smooth at x if and only if $dg_x: T_x X \rightarrow T_y Y$ is surjective. No smoothness of Y is assumed.*

Proof. If g is smooth at x , the infinitesimal lifting property lifts every tangent vector at y to one at x . Conversely, assume that dg_x is surjective, and let $e = \dim T_y Y$ and $N = \dim_x X$. Choose a surjection $\mathbb{C}[[y_1, \dots, y_e]] \rightarrow \hat{\mathcal{O}}_{Y, y}$ with kernel $I \subset (y_1, \dots, y_e)^2$. The cotangent map is injective, so the images of the y_i extend to a regular system of parameters of $\hat{\mathcal{O}}_{X, x} \cong \mathbb{C}[[x_1, \dots, x_N]]$; take $x_i = y_i$ for $i \leq e$. The composite $\mathbb{C}[[y_1, \dots, y_e]] \rightarrow \hat{\mathcal{O}}_{Y, y} \rightarrow \hat{\mathcal{O}}_{X, x}$ is then injective, and it kills I ; hence $I = 0$. Thus $\hat{\mathcal{O}}_{X, x}$ is a power series ring over $\hat{\mathcal{O}}_{Y, y} \cong \mathbb{C}[[y_1, \dots, y_e]]$. It is flat over $\mathcal{O}_{Y, y}$, and since $\mathcal{O}_{X, x} \rightarrow \hat{\mathcal{O}}_{X, x}$ is faithfully flat, $\mathcal{O}_{X, x}$ is flat over $\mathcal{O}_{Y, y}$. The fibre through x has completed local ring $\mathbb{C}[[x_{e+1}, \dots, x_N]]$, so it is smooth at x . A morphism that is flat at x with fibre smooth at x is smooth at x [12, Théorème 17.5.1]. $\square$

The smoothness of X cannot be omitted. The closed immersion

$$\mathrm{Spec} \mathbb{C}[t]/(t^2) \longrightarrow \mathrm{Spec} \mathbb{C}[t]/(t^3)$$

is an isomorphism on tangent spaces, but it is not flat.

Theorem 4.13 (The smooth locus of the centre morphism). *Let U be as in Theorem 4.6 and $B \in U(\mathbb{C})$. The centre morphism is smooth at B if and only if the centre tangent map*

$$\chi_B: HH^2(B) \longrightarrow T_{\mathrm{com}}^1(Z(B))$$

is surjective. These points form the $\mathbb{C}$ -points of a dense open subscheme $U^{\mathrm{cent}} \subset U$, which contains every semisimple algebra in U .

Proof. The torsor $\mathcal{F}_U$ is locally trivial with smooth fibre G_c over the smooth scheme U , so it is smooth. Lemma 4.12 and Proposition 4.11 give the criterion. The smooth locus of cent is open and, by equivariance, G_c -stable. Its image in U is therefore open, and it is the subscheme U^{cent} . If B is semisimple, then $HH^2(B) = 0$, and $Z(B) \cong \mathbb{C}^c$ has $T_{\mathrm{com}}^1 = 0$ by Lemma 4.2; so χ_B is surjective. The semisimple locus is dense in U . $\square$

Theorem 4.13 describes the smooth locus by the surjectivity of χ_B . The next subsection shows that this locus need not contain the original contraction algebra A , even though its centre has constant dimension on U .

4.5 A counterexample in type D

For $n \geq 2$, consider the algebra $D_{n,2}$ of [16, Definition 3.2 and Proposition 3.3],

$$A_n := D_{n,2} = \frac{\mathbb{C}\langle x, y \rangle}{(xy + yx, y^2 + x^2 + x^{2n-1}, x^{2n+1})}. \quad (4.5)$$

It is the Jacobi algebra of the potential $\frac{1}{2n}x^{2n} + \frac{1}{3}x^3 + xy^2$; the ideal contains a power of (x, y) , so completion changes nothing [16, Proposition 3.3]. By Brown–Wemyss, A_n is the contraction algebra of a simple flop of a smooth threefold, of type D and length two [4, Theorems 1.5 and 8.11]. It has dimension $4n$ and can only deform to $M_2(\mathbb{C})^{\times(n-1)} \times \mathbb{C}^4$ [16, Theorems 3.7 and 3.13]. By Theorem 3.1, its GV invariants are therefore $n_1 = 4$ and $n_2 = n - 1$, and $c = n + 3$. Thus A_n lies in the geometric setting of our theorems.

Throughout this subsection we write

$$\mathbf{u} = x^2, \quad \mathbf{v} = x^{2n-1}, \quad \mathbf{w} = x^{2n-2}y,$$

in sans-serif letters to distinguish them from the geometric parameters u and w of Sections 2 and 3.

Lemma 4.14 (Structure of $D_{n,2}$). *Let $n \geq 2$.*

- (i) *In A_n , $x^{2n-1}y = 0$, and the elements $1, x, \dots, x^{2n}$ and $y, xy, \dots, x^{2n-2}y$ form a basis. With $x^i = 0$ for $i > 2n$ and $x^i y = 0$ for $i > 2n - 2$,*

$$x^i y \cdot x^j = (-1)^j x^{i+j} y, \quad x^i y \cdot x^j y = (-1)^{j+1} (x^{i+j+2} + x^{i+j+2n-1}).$$

(ii) The centralizer of x is $\mathbb{C}[x] \oplus \mathbb{C}\mathbf{w}$, where $\mathbb{C}[x] = \text{span}(1, x, \dots, x^{2n})$.

(iii) The centre is

$$Z_n := Z(A_n) = \text{span}(1, \mathbf{u}, \dots, \mathbf{u}^n, \mathbf{v}, \mathbf{w}) \cong \frac{\mathbb{C}[\mathbf{u}, \mathbf{v}, \mathbf{w}]}{(\mathbf{u}^{n+1}, \mathbf{u}\mathbf{v}, \mathbf{u}\mathbf{w}, \mathbf{v}^2, \mathbf{v}\mathbf{w}, \mathbf{w}^2)},$$

of dimension $n + 3$.

(iv) $\mathbf{w}x = x\mathbf{w} = 0$, and $\mathbf{v}A_n + A_n\mathbf{v} \subseteq \mathbb{C}[x]$.

Proof. (i) Since $yx^i = (-1)^i x^i y$, multiplying $y^2 = -x^2 - x^{2n-1}$ by y on the left and on the right gives $-x^2 y + x^{2n-1} y = -x^2 y - x^{2n-1} y$, so $x^{2n-1} y = 0$. The listed elements span, and they are linearly independent by [16, Proposition 3.3]. The products follow from $yx^j = (-1)^j x^j y$ and the relation for y^2 .

(ii) Write $a = f(x) + g(x)y$. Then $[a, x] = [g(x)y, x] = -2xg(x)y$. By (i), this vanishes exactly when g is a multiple of x^{2n-2} .

(iii) By (ii), a central element has the form $f(x) + \lambda \mathbf{w}$. Since $2n - 2$ is even, $\mathbf{w}$ commutes with y . Moreover $[f(x), y] = 2 \sum_{i \text{ odd}} f_i x^i y$, which vanishes exactly when $f_i = 0$ for all odd $i \leq 2n - 3$. This gives the displayed basis. The relations hold because $x^{2n+1} = 0$, $x^{2n-1} y = 0$ and $\mathbf{w}^2 = x^{4n-4} y^2 = -x^{4n-2} - x^{6n-5} = 0$; comparing dimensions gives the presentation.

(iv) This follows from the products in (i). $\square$

Lemma 4.15 (A polynomial family of centres). *Let*

$$\mathcal{Z}_n = \frac{\mathbb{C}[t, \mathbf{u}, \mathbf{v}, \mathbf{w}]}{(\mathbf{u}^{n+1}, \mathbf{u}\mathbf{v}, \mathbf{u}\mathbf{w}, \mathbf{v}\mathbf{w}, \mathbf{w}^2, \mathbf{v}^2 - t\mathbf{w})}. \quad (4.6)$$

(i) The algebra $\mathcal{Z}_n$ is free over $\mathbb{C}[t]$ with basis $1, \mathbf{u}, \dots, \mathbf{u}^n, \mathbf{v}, \mathbf{w}$, and $\mathcal{Z}_n/(t) \cong Z_n$.

(ii) For $t \neq 0$, the fibre is isomorphic to $\mathbb{C}[\mathbf{u}, \mathbf{v}]/(\mathbf{u}^{n+1}, \mathbf{u}\mathbf{v}, \mathbf{v}^3)$, which is smoothable.

(iii) In the basis of (i), $\mathcal{Z}_n$ defines a morphism $\mathbb{A}^1 \rightarrow \text{Com}_{n+3}$ that factors through Sm_{n+3} . Its tangent vector at $t = 0$ is the cocycle $\psi_0 \in Z_{\text{com}}^2(Z_n)$ with $\psi_0(\mathbf{v}, \mathbf{v}) = \mathbf{w}$ and $\psi_0 = 0$ on all other pairs of basis vectors.

Proof. (i) On the free $\mathbb{C}[t]$ -module with the stated basis, let the powers of $\mathbf{u}$ multiply as in $\mathbb{C}[\mathbf{u}]/(\mathbf{u}^{n+1})$, put $\mathbf{v}^2 = t\mathbf{w}$, and let all other products of two basis vectors other than 1 vanish. The element $\mathbf{w}$ annihilates $\mathbf{u}, \dots, \mathbf{u}^n, \mathbf{v}, \mathbf{w}$, and $\mathbf{v}$ annihilates the powers of $\mathbf{u}$. Hence every product of three such basis vectors is zero in either order, unless all three are powers of $\mathbf{u}$. So the multiplication is associative and commutative. It satisfies the relations of (4.6), and those relations reduce every monomial to the basis; this gives inverse isomorphisms. Setting $t = 0$ recovers Lemma 4.14(iii).

(ii) For $t \neq 0$, eliminate $\mathbf{w} = t^{-1}\mathbf{v}^2$. The family

$$\frac{\mathbb{C}[h, \mathbf{u}, \mathbf{v}]}{(\mathbf{u}\mathbf{v}, \mathbf{u}^{n+1} - h^n \mathbf{u}, \mathbf{v}^3 - h^2 \mathbf{v})}$$

is free over $\mathbb{C}[h]$ with basis $1, \mathbf{u}, \dots, \mathbf{u}^n, \mathbf{v}, \mathbf{v}^2$. Indeed, it maps onto the fibre product over $\mathbb{C}[h]$ of $\mathbb{C}[h][\mathbf{u}]/(\mathbf{u}^{n+1} - h^n \mathbf{u})$ and $\mathbb{C}[h][\mathbf{v}]/(\mathbf{v}^3 - h^2 \mathbf{v})$ along $\mathbf{u} \mapsto 0$ and $\mathbf{v} \mapsto 0$; that fibre product is

free with the images of these elements as a basis, and they span the source. For $h \neq 0$, its fibre has the $n + 3$ distinct points $(u, v) = (h\xi, 0)$ with $\xi^n = 1$, and $(0, 0)$, $(0, h)$, $(0, -h)$, so it is isomorphic to $\mathbb{C}^{n+3}$. In this basis the family is a morphism $\mathbb{A}^1 \rightarrow \text{Com}_{n+3}$ mapping $h \neq 0$ into the irreducible set $G_{n+3} \cdot \mathbb{C}^{n+3}$. Its fibre at $h = 0$, which is the algebra in (ii), therefore lies in the closure Sm_{n+3} .

(iii) By (ii), the points with $t \neq 0$ map to Sm_{n+3} , which is closed; since $\mathbb{A}^1$ is reduced, the whole morphism factors through Sm_{n+3} . The structure constants are affine-linear in t , and the only t -dependent one is that of $v \cdot v$. $\square$

Theorem 4.16 (A centre deformation that does not lift). *Let $n \geq 2$, let $R_\varepsilon = \mathbb{C}[\varepsilon]/(\varepsilon^2)$, and let $\mathcal{Z}_{n,\varepsilon} = \mathcal{Z}_n \otimes_{\mathbb{C}[t]} R_\varepsilon$ with $t \mapsto \varepsilon$. There is no finite free R_ε -algebra $\mathcal{B}$ with $\mathcal{B}/\varepsilon\mathcal{B} \cong A_n$ together with a homomorphism of R_ε -algebras $\iota: \mathcal{Z}_{n,\varepsilon} \rightarrow \mathcal{B}$ whose reduction modulo ε is the inclusion $Z_n \subset A_n$.*

The image of ι is not required to be central, and no relation of A_n is required to hold in $\mathcal{B}$ beyond its reduction modulo ε .

Proof. Suppose that $(\mathcal{B}, \iota)$ exists. Since $\mathcal{B}$ is free over R_ε , multiplication by ε induces an isomorphism of A_n -bimodules

$$A_n \xrightarrow{\sim} \varepsilon\mathcal{B}, \quad a \longmapsto \varepsilon\tilde{a}, \quad (4.7)$$

where $\tilde{a}$ is any lift of a . We write εa for the image of a . Choose a lift $\tilde{x} \in \mathcal{B}$ of x , and put $\tilde{v} = \iota(v)$ and $\tilde{w} = \iota(w)$.

Since $x^{2n+1} = 0$, we have $\tilde{x}^{2n+1} = \varepsilon b$ for some $b \in A_n$. As $\tilde{x}$ commutes with its own powers, $0 = [\tilde{x}^{2n+1}, \tilde{x}] = \varepsilon[b, x]$, so $[b, x] = 0$ by (4.7). Lemma 4.14(ii) gives $b \in \mathbb{C}[x] \oplus \mathbb{C}w$.

Next, $\tilde{v}$ lifts $v = x^{2n-1}$, so $\tilde{v} = \tilde{x}^{2n-1} + \varepsilon g$ for some $g \in A_n$. Since $\varepsilon^2 = 0$ and $4n - 2 \geq 2n + 1$,

$$\tilde{v}^2 = \tilde{x}^{2n+1}\tilde{x}^{2n-3} + \varepsilon(vg + gv) = \varepsilon(bx^{2n-3} + vg + gv).$$

Here $2n - 3 \geq 1$ and $wx = 0$, so $bx^{2n-3} \in \mathbb{C}[x]$, and $vg + gv \in \mathbb{C}[x]$ by Lemma 4.14(iv). Hence $\tilde{v}^2 \in \varepsilon\mathbb{C}[x]$. On the other hand, $\tilde{v}^2 = \iota(v^2) = \iota(\varepsilon w) = \varepsilon\tilde{w} = \varepsilon w$. By (4.7), $w \in \mathbb{C}[x]$, which contradicts Lemma 4.14(i). $\square$

Corollary 4.17 (The centre morphism is not smooth at $D_{n,2}$). *Let $n \geq 2$, $c = n + 3$, and let U be a neighbourhood of A_n as in Theorem 4.6.*

- (i) *The class $[\psi_0]$ is not in the image of χ_{A_n} . Hence the centre morphism is not smooth at A_n , and $A_n \notin U^{\text{cent}}$.*
- (ii) *The centre morphism factors through Sm_c , which is the scheme-theoretic image of $\text{cent}: \mathcal{F}_U \rightarrow \text{Com}_c$. The morphism $\text{cent}: \mathcal{F}_U \rightarrow \text{Sm}_c$ is not smooth at the points over A_n either.*
- (iii) *The family $\mathcal{Z}_n$ does not lift near $t = 0$: there is no finite free algebra over $\mathbb{C}[t]_{(t)}$, over the convergent power series in t , or over $\mathbb{C}[[t]]$, with special fibre A_n and centre identified with $\mathcal{Z}_n$ compatibly with $Z_n \subset A_n$.*

In particular, the second half of Conjecture 4.1(2) fails for A_n , although the first half holds by Theorem 4.6.

Proof. (i) Suppose that $[\psi_0] = \chi_{A_n}([\phi])$. By Lemma 4.10, there is a map τ for which (4.3) equals ψ_0 , after changing τ by an element of $C_{\text{norm}}^1(Z_n, Z_n)$. The map $z \mapsto z + \varepsilon\tau(z)$ is then an algebra homomorphism $\mathcal{Z}_{n,\varepsilon} \rightarrow (A_n)_\phi$ lifting $Z_n \subset A_n$, contradicting Theorem 4.16. The remaining assertions follow from Theorem 4.13.

(ii) Over the dense semisimple locus of U , the centres are isomorphic to $\mathbb{C}^c$, so cent maps a dense open subscheme of the reduced scheme $\mathcal{F}_U$ into $G_c \cdot \mathbb{C}^c$. Hence cent factors through Sm_c . By equivariance its image contains the open orbit, so the scheme-theoretic image is Sm_c . Let $\mathfrak{f}$ be the frame of Z_n given by the basis of Lemma 4.15(i). By part (iii) of that lemma, ψ_0 is tangent to Sm_c at $\text{cent}(A_n, \mathfrak{f})$. By (i) and Proposition 4.11, it is not in the image of $d\text{cent}_{(A_n, \mathfrak{f})}$. So the differential of $\text{cent}: \mathcal{F}_U \rightarrow \text{Sm}_c$ at $(A_n, \mathfrak{f})$ is not surjective, and cent is not smooth there by Lemma 4.12; by equivariance, the same holds at every point over A_n .

(iii) Reducing such a lift modulo t^2 would give a pair $(\mathcal{B}, \iota)$ as excluded by Theorem 4.16. $\square$

Remark 4.18. The socle element $\mathbf{v} = x^{2n-1}$ of Z_n is a power of the noncentral element x . In every first-order deformation, the square of a lift of $\mathbf{v}$ stays in the subalgebra generated by x , because of the relation $x^{2n+1} = 0$ and the shape of the centralizer of x . The commutative deformation $\mathbf{v}^2 = tw$ of the centre forgets this, and asks the square to acquire the component $\mathbf{w} = x^{2n-2}y$.

The obstruction is already visible to first order. It does not exclude a lift of $\mathcal{Z}_n$ after a ramified base change $t = h^r$ with $r \geq 2$, in which the first-order part vanishes; we make no claim about such lifts.

4.6 Exact tangent data for $D_{2,2}$

For $n = 2$, the tangent spaces can be computed completely. Here $A = A_2$ has basis $1, x, x^2, x^3, x^4, y, xy, x^2y$, and Z_2 has basis $1, \mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{q}$ with $\mathbf{u} = x^2$, $\mathbf{v} = x^3$, $\mathbf{w} = x^2y$ and $\mathbf{q} = \mathbf{u}^2 = x^4$. The only nonzero product of two of $\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{q}$ is $\mathbf{u}^2 = \mathbf{q}$.

Proposition 4.19 (Tangent data of $D_{2,2}$). *For $A = D_{2,2}$ and $Z_2 = Z(A)$, the following hold.*

(i) $\dim \text{Der}(A) = 6$, $\dim HH^1(A) = \dim HH^2(A) = 3$, and $\dim Z_{\text{norm}}^2(A, A) = 53 = \dim_A X_8$.

(ii) $\dim B_{\text{com}}^2(Z_2) = 10$, $\dim Z_{\text{com}}^2(Z_2) = 26$, and $\dim T_{\text{com}}^1(Z_2) = 16$.

(iii) *The centre tangent map χ_A has rank 3. Its image is spanned by the classes of the cocycles ψ_1, ψ_2, ψ_3 whose values on pairs of basis vectors other than 1 vanish, except for*

$$\begin{aligned} \psi_1(\mathbf{u}, \mathbf{w}) &= -\mathbf{w}, & \psi_1(\mathbf{u}, \mathbf{q}) &= -2\mathbf{q}, & \psi_1(\mathbf{v}, \mathbf{v}) &= \mathbf{q}; \\ \psi_2(\mathbf{u}, \mathbf{v}) &= -\tfrac{1}{2}\mathbf{w}, & \psi_2(\mathbf{u}, \mathbf{w}) &= \tfrac{1}{2}\mathbf{v}, & \psi_2(\mathbf{v}, \mathbf{w}) &= \tfrac{1}{2}\mathbf{q}; \\ \psi_3(\mathbf{u}, \mathbf{w}) &= -2\mathbf{w}, & \psi_3(\mathbf{u}, \mathbf{q}) &= -2\mathbf{q}, & \psi_3(\mathbf{w}, \mathbf{w}) &= 2\mathbf{q}, \end{aligned}$$

together with the values forced by symmetry.

(iv) *At every point of $\mathcal{F}_U$ over A , $\text{rank } d\text{cent} = 10 + 3 = 13$, whereas $\dim T_{Z_2} \text{Sm}_5 \geq \dim \text{Sm}_5 = 20$ and $\dim T_{Z_2} \text{Com}_5 = 26$.*

Proof. Parts (i)–(iii) are exact linear algebra over $\mathbb{Q}$, described in Appendix C. Part (iv) follows from (ii), (iii), Proposition 4.11, and Lemma 4.2. $\square$

Thus, at $D_{2,2}$, the differential of the centre morphism misses at least seven dimensions of the tangent space of its scheme-theoretic image.

Theorem 4.16 fixes the identification of the special fibre of $\mathcal{Z}_{n,\varepsilon}$ with Z_n . This marked obstruction suffices to disprove smoothness of the centre morphism. For $n = 2$, the nonlifting statement also holds without a chosen identification.

Proposition 4.20 (Non-lifting without a marking). *No first-order deformation of $A = D_{2,2}$ over R_ε has centre isomorphic to $\mathcal{Z}_{2,\varepsilon}$ as an R_ε -algebra. Equivalently, the $\text{Aut}(Z_2)$ -orbit of $[\psi_0]$ does not meet the image of χ_A .*

Proof. Let $\mathfrak{m} = (\mathfrak{u}, \mathfrak{v}, \mathfrak{w}, \mathfrak{q})$ be the maximal ideal of Z_2 . Then $\mathfrak{m}^2 = \mathbb{C}\mathfrak{q}$ and $\text{Soc}(Z_2) = \mathbb{C}\mathfrak{v} \oplus \mathbb{C}\mathfrak{w} \oplus \mathbb{C}\mathfrak{q}$. Consider the following linear condition on $\psi \in Z_{\text{com}}^2(Z_2)$:

$$\psi(s, s) \in \mathbb{C}s + \mathfrak{m}^2 \quad \text{for every } s \in \text{Soc}(Z_2). \quad (4.8)$$

It depends only on the class of ψ : for $s \in \text{Soc}(Z_2)$ we have $s^2 = 0$ and $\mathfrak{m}s = 0$, so $(\delta f)(s, s) = 2sf(s)$ is a multiple of s for every $f \in C_{\text{norm}}^1(Z_2, Z_2)$. It is invariant under $\text{Aut}(Z_2)$, which preserves $\text{Soc}(Z_2)$ and $\mathfrak{m}^2$. By Proposition 4.19(iii), the image of χ_A satisfies (4.8): indeed $\psi_i(s, s) \in \mathbb{C}\mathfrak{q}$ for all $s \in \text{Soc}(Z_2)$ and $i = 1, 2, 3$. The class $[\psi_0]$ violates it, since $\psi_0(\mathfrak{v}, \mathfrak{v}) = \mathfrak{w} \notin \mathbb{C}\mathfrak{v} + \mathbb{C}\mathfrak{q}$.

Now let $\mathcal{B}$ be a first-order deformation of A , and suppose that $Z(\mathcal{B}) \cong \mathcal{Z}_{2,\varepsilon}$. By Corollary 4.7, $Z(\mathcal{B})$ reduces to Z_2 , and with this marking it represents a class $\chi_A([\phi])$. The isomorphism reduces to an automorphism σ of Z_2 , and it identifies $[\psi_0]$ with the transport of $\chi_A([\phi])$ by σ . This contradicts the invariance of (4.8). $\square$

We do not claim the unmarked statement for $n > 2$.

4.7 Consequences for the conjecture

Corollary 4.21 (Conjecture 4.1 for simple flops). *Let A be the contraction algebra of a completed simple flop of a smooth complex threefold.*

- (1) *Conjecture 4.1(1) holds: A is a smooth point of X_d , with completed local ring (4.1), and A is unobstructed.*
- (2) *The first half of Conjecture 4.1(2) holds: near A the centre and the cocenter are dual vector bundles of rank $\sum_j n_j$, compatible with arbitrary base change. The second half fails in general: the centre morphism is smooth exactly where the centre tangent map is surjective, and this fails at $D_{n,2}$ for every $n \geq 2$, even with the smoothable component or the image as target.*

Proof. Combine Theorems 4.4, 4.6 and 4.13 with Corollary 4.17. $\square$

The conclusions concern three distinct deformation problems: extending an algebra multiplication over a larger base, forming centres in an existing family, and lifting a prescribed deformation of the centre. The first two have positive answers here; the counterexample concerns the third.

The corollary covers the contraction algebras of types A and D considered in [16], all of which arise from simple flops of smooth threefolds [4, Theorem 1.5]. It does not address

contraction algebras outside this setting, lifts of centre families after ramified base change, or the structure of the image of the deformation functor of A in that of $Z(A)$ beyond first order. In particular, the three-dimensional image of χ_A at $D_{2,2}$ does not by itself define a smooth subfunctor of the commutative deformations of Z_2 .

We now return to geometric families. The next section proves that the relative contraction quotient is finite free over an arbitrary complete Noetherian local base, allowing the centre and cocenter results above to apply to those families.

5 Flatness over arbitrary deformation bases

We now prove the algebra-flatness assertion needed in Remark 4.9. For a deformation of a completed simple flop over any complete Noetherian local complex base, the relative contraction quotient of the fixed central generator is finite free. Its centre and cocenter are then finite free and commute with base change. This is the corresponding case of [14, Conjecture 4.3]. Kawamata proved algebra flatness and special-fibre compatibility over a disk, allowing several exceptional curves [19, Theorem 3.1(2)].

The proof has an algebraic and a geometric part. First we construct a square free resolution of the universal marked quotient and prove a flatness criterion for its finite slices. We then identify an arbitrary completed geometric deformation with such a slice. Keeping the central generator fixed retains the matrix multiplicities when a fibre splits.

5.1 A square presentation of the universal quotient

Retain the rings $R_H \subset \tilde{R}$, the marked quotient $\mathcal{C}_H$, and the reduced curve-divisor equations P_j from Section 3.7.

Proposition 5.1 (The weighted discriminant presentation). *The finite R_H -module $\mathcal{C}_H$ has a free resolution*

$$0 \longrightarrow R_H^a \xrightarrow{\Theta_A} R_H^a \longrightarrow \mathcal{C}_H \longrightarrow 0 \quad (5.1)$$

with

$$\mathrm{Fitt}_0^{R_H}(\mathcal{C}_H) = (\det \Theta_A) = \left(\prod_{j=1}^{\ell} P_j^{j^2} \right). \quad (5.2)$$

In particular, $\mathcal{C}_H$ is a Cohen–Macaulay module of codimension one over R_H .

Proof. Use the central extension E_p of Section 3.5. Theorem 3.2(i) of [8] makes the whole universal algebra $\Pi[z]$ finite free over $K_\rho = \mathbb{C}[x_i - \rho_i z]$. Its retained corner after flat completion is a direct summand of a finite free $\tilde{R}$ -module, hence is free since $\tilde{R}$ is local. Multiplication by z is injective on $\Pi[z]$: before changing the coefficient ring, z is a polynomial variable. This injection survives flat completion and the corner projection. Setting $z = 0$ gives

$$0 \longrightarrow E_p \xrightarrow{z} E_p \longrightarrow \mathcal{D}_p \longrightarrow 0. \quad (5.3)$$

This argument uses the whole-algebra theorem and is valid for the coefficient-five E_8 vertex as well.

The generic decomposition (3.13) identifies $(E_p)_L$ with the product of $M_{m_p(\alpha)}(L)$, for positive roots with $m_p(\alpha) > 0$. On that factor z acts as $-(\lambda, \alpha)/(\rho, \alpha)$ [8, Theorem 3.2(v) and the following display]. Consequently, as an identity in the domain $\tilde{R}$,

$$\det(z \mid E_p) = \eta \prod_{\substack{\alpha \in \Phi^+ \\ m_p(\alpha) > 0}} (\lambda, \alpha)^{m_p(\alpha)^2}, \quad \eta \in \mathbb{C}^\times.$$

The determinant is nonzero, so $\mathcal{D}_p$ is torsion and has projective dimension one over $\tilde{R}$.

By (3.25), $\mathcal{C}_H \otimes_{R_H} \tilde{R} \cong \mathcal{D}_p$. Choose a finite free surjection $R_H^a \twoheadrightarrow \mathcal{C}_H$ with kernel K . The first syzygy $K \otimes_{R_H} \tilde{R}$ is projective by (5.3). Finite projectivity descends along the faithfully flat map $R_H \rightarrow \tilde{R}$, so K is free. Since $\mathcal{C}_H$ is torsion, its rank is zero and $\text{rk } K = a$. This proves (5.1).

Fitting ideals commute with scalar extension. The root-orbit and reduced-divisor identifications in Lemmas 2.5 and 2.6 identify the displayed determinant ideal with the extension of $(\prod_j P_j^{j^2})$. Faithful flatness then gives (5.2) downstairs. The regular local ring R_H and Auslander–Buchsbaum show that the nonzero torsion module $\mathcal{C}_H$, of projective dimension one, has depth and dimension $\dim R_H - 1$. $\square$

The exponent j^2 is the dimension of a matrix block as an algebra. On its cocenter the same block contributes one dimension; accordingly Theorem 3.16 has determinant $\prod_j P_j$. Both universal modules are supported on the curve discriminant. The next lemma shows how a family of slices produces flatness over the external deformation base.

5.2 Finite slices over a nonreduced base

Lemma 5.2 (Finite slices of square presentations). *Let $(B, \mathfrak{m}, k)$ be a complete Noetherian local ring, put $D_B = B[[w]]$, and let $\Phi : D_B^a \rightarrow D_B^a$ be a square matrix. If $\det \bar{\Phi} \neq 0$ in $k[[w]]$, then Φ is injective and its cokernel is finite free over B , of rank $\text{ord}_w \det \bar{\Phi}$.*

Proof. Put $M = \text{coker } \Phi$. Its closed fibre is finite-dimensional by Smith normal form over $k[[w]]$. Lift a basis to $m_1, \dots, m_b \in M$. Successive reduction modulo $\mathfrak{m}M$ writes every $m \in M$ as

$$m = \sum_i b_i^{(n)} m_i + r_n, \quad r_n \in \mathfrak{m}^n M, \quad b_i^{(n+1)} - b_i^{(n)} \in \mathfrak{m}^n.$$

The coefficients converge in B . Since M is finite over the Noetherian local ring D_B , Krull intersection gives $\bigcap_n \mathfrak{m}^n M = 0$. The limiting coefficients express m as a B -linear combination of the m_i . Thus M is finite over B .

If $\Phi v = 0$, injectivity of $\bar{\Phi}$ gives $v \in \mathfrak{m} D_B^a$. On each associated graded piece the map is

$$1 \otimes \bar{\Phi} : (\mathfrak{m}^n / \mathfrak{m}^{n+1}) \otimes_k k[[w]]^a \longrightarrow (\mathfrak{m}^n / \mathfrak{m}^{n+1}) \otimes_k k[[w]]^a,$$

which is injective. Induction and Krull intersection give $v = 0$. Here the graded-piece identification follows from the flatness of $B[[w]]$ over the Noetherian ring B .

The resulting exact sequence is a flat resolution over B , so

$$\text{Tor}_1^B(M, k) = \ker \bar{\Phi} = 0.$$

Take a minimal free surjection $B^b \rightarrow M$, with finite kernel L . The Tor vanishing and the isomorphism on reduction give $L/\mathfrak{m}L = 0$. Nakayama implies $L = 0$. Finally Smith normal form gives $b = \dim_k(M/\mathfrak{m}M) = \text{ord}_w \det \bar{\Phi}$. $\square$

Corollary 5.3 (Derived pullback along a finite slice). *Suppose a finite module C over a Noetherian local ring R has an exact sequence $0 \rightarrow R^a \xrightarrow{\Theta} R^a \rightarrow C \rightarrow 0$. For a local map $F : R \rightarrow B[[w]]$, if $C \otimes_R k[[w]]$ is finite-dimensional, then*

$$C \otimes_R^L B[[w]] \simeq (C \otimes_R B[[w]])[0].$$

The ordinary pullback is finite free over B , with rank equal to the dimension of that closed fibre.

Proof. The finite closed-fibre cokernel forces the square matrix $\overline{\Theta(F)}$ to be invertible over $k((w))$. Lemma 5.2 applies to $\Theta(F)$ and also shows that the pulled-back resolution has no negative cohomology. $\square$

5.3 Classifying an arbitrary geometric deformation

To apply the slice criterion to a given geometric family, we must identify its relative contraction quotient with a pullback of $\mathcal{C}_H$. Fix a completed simple flop $f_0 : X_0 \rightarrow Y_0$ of a smooth complex threefold. Write $\mathcal{T}_0 = \mathcal{O} \oplus \mathcal{T}'_0$ for its selected basic tilting generator and A_0 for its contraction algebra. A completed deformation over $(B, \mathfrak{m}, \mathbb{C})$ means a projective contraction $f_B : X_B \rightarrow Y_B$, with affine Noetherian target, source and target flat over B , and closed fibre f_0 . We work in the formal neighbourhood of the central contraction point and its inverse image. The distinguished structure-sheaf summand is retained in every comparison. These are deformations of the contraction, including its chosen small resolution, rather than deformations of the target singularity alone.

Proposition 5.4 (Families of classifying arcs). *Let B be complete Noetherian local with residue field $\mathbb{C}$, and let f_B be such a completed deformation. A lift of a general surface-section parameter supplies a continuous local map*

$$F : R_H \longrightarrow D_B = B[[w]]$$

and identifies f_B with the completed pullback of the universal marked partial resolution. There is a relative tilting generator $\mathcal{T}_B$ lifting $\mathcal{T}_0$. For any such lift with the fixed summands, its relative contraction quotient satisfies

$$\mathcal{A}_B := \frac{\mathrm{End}_{X_B}(\mathcal{T}_B)^{\mathrm{op}}}{\mathrm{End}_{X_B}(\mathcal{T}_B)^{\mathrm{op}} e_0 \mathrm{End}_{X_B}(\mathcal{T}_B)^{\mathrm{op}}} \cong \mathcal{C}_H \otimes_{R_H, F} D_B. \quad (5.4)$$

Its reduction modulo $\mathfrak{m}$ is A_0 .

Proof. Choose a general section parameter w_0 on the central target and lift it to w on the completed target. It is topologically nilpotent in the local target ring, so this defines the action of $B[[w]]$. On the central source and target, w_0 is a non-zero-divisor: the target is a cDV domain and the source is smooth and integral. Their local rings are therefore torsion-free, hence flat, over $\mathbb{C}[[w]]$. Apply the fibrewise local criterion for flatness to $B \rightarrow D_B \rightarrow Q$, where Q is a local ring of the completed source or target. The module Q is flat over B , and $Q/\mathfrak{m}Q$ is flat over $D_B/\mathfrak{m}D_B$. Thus Q is flat over D_B [25, Tag 00MP].

The fibre over $(\mathfrak{m}, w)$ is the marked partial resolution of the ADE surface section. The formal versality statement of [17, Section 2.3] applies to arbitrary formal deformations of this surface, including those over nonreduced bases. Its base is $H = \mathfrak{h}/W_0$. Applying versality successively

to Artin quotients of D_B , compatibly with the preceding choices, gives $F : R_H \rightarrow D_B$ and the identification of the formal sources. The contraction targets agree by relative affinization. More explicitly, $Rf_{0*}\mathcal{O}_{X_0} = \mathcal{O}_{Y_0}$; on a square-zero thickening, flatness of source and target identifies the kernel sheaves with the central structure sheaves tensored by the square-zero ideal. Proper pushforward and the central higher-direct-image vanishing then give the same equality on the thickening. Induction and formal functions give $Rf_{B*}\mathcal{O}_{X_B} = \mathcal{O}_{Y_B}$. This also identifies the morphism to the affine target.

The universal tilting generator pulls back by Lemmas 2.19 and 2.20. Its central restriction is $\mathcal{T}_0$, with its specified summands, by Proposition 2.22 and the same rank-and-determinant argument on the central arc. Any other lift is isomorphic to it: vanishing of the first and second self-Ext groups of $\mathcal{T}_0$ lifts the summands and their isomorphisms through each square-zero thickening. The compatible isomorphisms pass to the formal limit, as in Proposition 3.15; they preserve e_0 . Ordinary endomorphism base change (Proposition 2.21) and right exactness of the idempotent quotient now give (5.4).

The reduction F_0 modulo $\mathfrak{m}$ classifies the original flop. Lemma 3.17 identifies its quotient with A_0 . In particular it is finite-dimensional, so Proposition 5.1 and Corollary 5.3 make the right side of (5.4) finite over B . Its w -action is nilpotent modulo $\mathfrak{m}$; on this finite module the $(\mathfrak{m}, w)$ -adic and $\mathfrak{m}$ -adic topologies are equivalent. Thus the completed and ordinary quotient expressions agree, with their full finite relative support retained. $\square$

5.4 The Hua–Toda flatness theorem

Theorem 5.5 (Arbitrary-base Hua–Toda flatness for simple flops). *Let f_0 and f_B be as in Proposition 5.4. Put*

$$d = \dim_{\mathbb{C}} A_0 = \sum_j j^2 n_j, \quad N = \sum_j n_j.$$

Then $\mathcal{A}_B$ is finite free over B of rank d , and $HH_0(\mathcal{A}_B/B)$ and $Z(\mathcal{A}_B)$ are finite free of rank N . For every commutative B -algebra B' , the canonical maps

$$\begin{aligned} HH_0(\mathcal{A}_B/B) \otimes_B B' &\xrightarrow{\sim} HH_0(\mathcal{A}_B \otimes_B B'/B'), \\ Z(\mathcal{A}_B) \otimes_B B' &\xrightarrow{\sim} Z(\mathcal{A}_B \otimes_B B') \end{aligned}$$

are isomorphisms. Moreover

$$\mathcal{C}_H \otimes_{R_{H,F}}^L B[[w]] \simeq \mathcal{A}_B[0].$$

No reducedness, regularity, or dimension assumption on B , and no semisimplicity assumption on a generic fibre, is required.

Proof. Pull back (5.1) along F . The closed-fibre cokernel is A_0 , so Corollary 5.3 gives algebra freeness, rank d , and the derived statement. In terms of the divisor equations,

$$d = \text{ord}_w \det \Theta_A(F_0) = \sum_j j^2 \text{ord}_w P_j(F_0) = \sum_j j^2 n_j,$$

where the last equality is Proposition 2.16. Flatness was obtained before using this numerical identification.

The centre and cocenter assertions now follow from Theorem 4.6. Indeed, choose a B -basis of $\mathcal{A}_B$ beginning with its identity and reducing to a fixed basis of A_0 . The resulting morphism

$\mathrm{Spec} B \rightarrow X_d$ factors through the open neighbourhood U of A_0 : an open subset of a local spectrum containing its closed point is the whole spectrum. Pulling back the two bundles and their universal base-change isomorphisms proves the claims for every B' .

For completeness, the universal cocenter presentation gives a second proof of the centre and cocenter assertions. Using (3.26) and the commutator-cokernel description of degree-zero Hochschild homology, we obtain

$$HH_0(\mathcal{A}_B/B) \cong \mathrm{coker}(\Theta(F) : D_B^q \rightarrow D_B^q).$$

The commutator submodule is unchanged on regarding the algebra as a B -algebra instead of a D_B -algebra, since $r[a, b] = [ra, b]$ for a central scalar r . Lemma 5.2 and $\mathrm{ord}_w \prod_j P_j(F_0) = N$ make the cocenter free. Lift a symmetrizing trace of A_0 through this free cocenter. The resulting trace $\tau_B : \mathcal{A}_B \rightarrow B$ has perfect pairing $(a, b) \mapsto \tau_B(ab)$, whose determinant reduces to a nonzero scalar. The proof of Lemma 3.10 over B then gives

$$Z(\mathcal{A}_B) \cong \mathrm{Hom}_B(HH_0(\mathcal{A}_B/B), B).$$

The perfect pairing persists after every scalar extension, and the dual of a finite free module commutes with base change. This proves the canonical centre comparison as well. $\square$

Corollary 5.6 (Conservation of the root schemes). *For each j , the scheme*

$$\mathrm{Spec}(B[[w]]/(P_j(F))) \longrightarrow \mathrm{Spec} B$$

is finite flat of degree n_j .

Proof. Apply Lemma 5.2 to the one-by-one matrix $P_j(F)$, whose closed specialization has order n_j . $\square$

Corollary 5.7 (Local algebraic and analytic bases). *Let a Noetherian algebraic or complex analytic family of contractions have a fibre whose completion is a simple flop of a smooth complex threefold. Suppose its relative contraction quotient, formed using a lift of the fixed central generator, is finite over a neighbourhood of that point, and use the completed contraction along its full finite relative support. Then near that point the quotient, centre, and cocenter are locally free of ranks d, N, N , and the latter two commute with arbitrary base change.*

Proof. Complete the local base ring. Endomorphism and quotient base change identify the completed relative algebra with that in Theorem 5.5. Finite local freeness descends along the faithfully flat completion. After shrinking, a local frame of the resulting algebra family maps into the neighbourhood U of Theorem 4.6; that theorem supplies both remaining assertions. $\square$

Remark 5.8 (The relative support and the scope of the conclusion). The theorem uses the relative idempotent quotient of a fixed generator. For non-Gorenstein external bases it does not identify that quotient with an endomorphism group in the singularity category of the total space without additional hypotheses. It proves the three flatness assertions of [14, Conjecture 4.3] in the completed simple-flop setting with this relative-algebra convention.

Base change is performed on the finite B -algebra, preserving all moving support points. For example, $B[[w]]/(w^2 - t)$, with $B = \mathbb{C}[[t]]$, is free of rank two over B , whereas $\mathbb{C}((t))[[w]]/(w^2 - t) = 0$. Completing anew at $w = 0$ after inverting t would discard the moving support. Similarly,

taking a basic Morita representative anew in each split fibre would replace a block M_j by $\mathbb{C}$ and lose its multiplicity j^2 . Neither operation is part of the relative construction. Finally, centre flatness describes the centres of the given deformation; it does not imply smoothness of the centre morphism or lifting of arbitrary commutative centre deformations.

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A Root enumeration for the primitive marked pairs

This appendix gives the finite calculation used in Lemma 2.5. Number the vertices in the diagrams of Figure 2 from left to right along the horizontal chain, and number the lower vertex last. Thus the retained vertices are 1, 2, 3, 3, 4, 3 for lengths 1, $\dots$, 6, respectively. This diagram numbering is used only for the enumeration; the table in the proof of Theorem 3.13 records Karmazyn's numbering.

Represent a root by its vector $v = (v_1, \dots, v_r)$ in the simple-root basis. If $i \sim h$ denotes adjacency in the diagram, then

$$s_i(v)_h = v_h \quad (h \neq i), \quad s_i(v)_i = -v_i + \sum_{h \sim i} v_h.$$

Starting from the simple roots, repeatedly apply all s_i and keep each new vector once. The process terminates because these are finite ADE root systems; by definition of the Weyl action, its output is the full root system. Retain the vectors with nonnegative coordinates to obtain Φ^+ .

For each marked coefficient $j > 0$, start with one vector satisfying $v_p = j$, and repeat the same procedure using only the reflections with $i \neq p$. Compare the resulting orbit with the complete set of positive roots of coefficient j . In all six cases they agree. The numbers of positive roots are

$$1, \quad 12, \quad 36, \quad 63, \quad 120, \quad 120,$$

and the non-Levi totals are

$$N_p = 1, \quad 9, \quad 29, \quad 53, \quad 104, \quad 106.$$

The individual orbit cardinalities are those of Lemma 2.5. The calculation also checks that the largest marked coefficient is the stated length.

The integer implementation is included in `tools/check_mathematics.py` in the source distribution. At each step it verifies

$$2 \sum_i v_i^2 - 2 \sum_{\{i,h\} \text{ an edge}} v_i v_h = 2.$$

All vector operations and comparisons are exact. Distinct coefficient orbits give disjoint sets of hyperplanes, since proportional roots in a reduced root system differ only by sign. This also verifies that the descended curve divisors in the lemma are distinct.

B The coefficient-five E_8 calculation

B.1 The ρ -line relations

We derive the relations of Section 3.6 directly from the preprojective relations. The algebra Π_0^ρ is the path algebra over $\mathbb{C}[z]$ of the double of a finite E_8 quiver Q , with z central, modulo

$$\sum_{a \in Q_1} [a, a^*] = z \sum_i e_i.$$

Up to isomorphism it does not depend on the orientation of Q [9, Section 2.1]. We compose paths from left to right. For an edge joining v and v' , let $\ell_{v,v'}$ be the summand of the left-hand side that is a loop at v through v' ; it is aa^* or $-a^*a$, according to the orientation of the arrow a of Q joining them. The relation at v reads $\sum_{v' \sim v} \ell_{v,v'} = z e_v$. If $p: v \rightarrow v'$ and $q: v' \rightarrow v$ are the two arrows of the double quiver, then $\ell_{v,v'} = \pm pq$ and $\ell_{v',v} = \mp qp$; hence

$$\ell_{v,v'} p = -p \ell_{v',v}, \quad q \ell_{v,v'} = -\ell_{v',v} q. \quad (\text{B.1})$$

Lemma B.1 (Arms). *Let $v_0, v_1, \dots, v_r$ be a chain in the Dynkin graph such that $v_1, \dots, v_r$ have no neighbours other than those in the chain. Then $X = \ell_{v_0, v_1}$ satisfies*

$$X(X+z)(X+2z) \cdots (X+rz) = 0.$$

Proof. Put $Y = \ell_{v_1, v_0}$. By (B.1), $qf(X) = f(-Y)q$ for every polynomial f over $\mathbb{C}[z]$, and therefore $Xf(X) = \pm pf(-Y)q$. It suffices to find f of degree r with $f(-Y) = 0$. If $r = 1$, the relation at v_1 is $Y = z$, and $f(T) = T + z$ works. If $r > 1$, the relation at v_1 gives $\ell_{v_1, v_2} = z - Y$, and by induction $\prod_{m=0}^{r-1} (\ell_{v_1, v_2} + mz) = 0$. Thus $\prod_{m=1}^r (mz - Y) = 0$, and $f(T) = \prod_{m=1}^r (T + mz)$ works. $\square$

The nodal corners of Etingof–Latour–Rains are presented by relations of the same form [9, equation (3.1)]. We number the vertices of E_8 as in [17, Example 4.26]: the finite diagram is the chain 1, 2, 3, 4, 8, 7, 6 with 5 attached to the branching vertex 8, and the retained vertex is $p = 4$. Choose Q so that its arrow a between 4 and 8 points from 4 to 8, and put

$$d = -\ell_{4,3}, \quad b = -\ell_{8,5}, \quad c = -\ell_{8,7}.$$

Lemma B.1, applied to the arms of lengths 3, 1 and 2, and the relations at the vertices 4 and 8 give

$$\begin{aligned} d(d-z)(d-2z)(d-3z) &= 0, & b(b-z) &= 0, & c(c-z)(c-2z) &= 0, \\ aa^* &= d + ze_4, & a^*a + b + c + ze_8 &= 0. \end{aligned} \quad (\text{B.2})$$

These are the relations of Karmazyn's intermediate presentation [17, Example 4.26] at $t_i = z$, after the free vertex has been removed. Put

$$B = b - 2ze_8, \quad C = c, \quad H = a^*a = -B - C - 3ze_8,$$

so that

$$B^2 = -3zB - 2z^2, \quad C^3 = 3zC^2 - 2z^2C. \quad (\text{B.3})$$

In the corner $B_5 = e_4 \Pi_0^\rho e_4$, define

$$\beta = aa^* + 3ze_4 = d + 4ze_4, \quad \gamma = aBa^* - 2z^2e_4.$$

Lemma B.2 (Generators). *The elements z, β, γ generate the algebra B_5 .*

Proof. A closed path at 4 is a product of closed paths that meet 4 only at their ends. Those entering the arm $\{1, 2, 3\}$ are polynomials in d and z , by the relations along the arm and (B.1), as in the proof of Lemma B.1. The others have the form $aq a^*$ with q a closed path at 8. In the same way, a closed path at 8 is a product of polynomials in b and z , polynomials in c and z , and elements $a^* r a$ with r a closed path at 4. Since $da = a(H - z)$ by (B.2), we have $a^* d^k a = H(H - z)^k$. Induction on length therefore shows that every closed path at 8 is a polynomial in B , C and z . Finally, substitute $C = -a^* a - B - 3z$ in an element $aq(B, C)a^*$. Each occurrence of $a^* a$ splits the expression into a product of two elements of the same form, and powers of B reduce by (B.3). Hence B_5 is generated by z, aa^* and aBa^* , that is, by z, β and γ . $\square$

Lemma B.3 (Relations). *In B_5 , the elements*

$$\begin{aligned} P &= (\beta - 4z)(\beta - 5z)(\beta - 6z)(\beta - 7z), \\ R_4 &= (\gamma + \beta^2)^2 + \beta\gamma\beta - 5z^2(\gamma + \beta^2) + 4z^4, \\ R_5 &= \gamma\beta\gamma + \gamma^2\beta + \gamma\beta^3 - 5z^2\gamma\beta \end{aligned}$$

vanish.

Proof. Since $\beta = d + 4z$, the element P equals $d(d - z)(d - 2z)(d - 3z) = 0$. For the others, multiply elements of the form $s e_4 + a P a^*$, with $s \in \mathbb{C}[z]$ and P a noncommutative polynomial in B and C , by the rule

$$(s e_4 + a P a^*)(s' e_4 + a P' a^*) = ss' e_4 + a(s P' + s' P + P H P') a^*.$$

Only (B.3) and $H = -B - C - 3z$ are used. They give $HB = 2z^2 - CB$, $BH = 2z^2 - BC$ and

$$\gamma + \beta^2 = 7z^2 + a(3z - C)a^*, \quad \beta\gamma = -6z^3 + a(3z - C)Ba^*, \quad \gamma\beta = -6z^3 + aB(3z - C)a^*.$$

For R_5 , these give $\beta\gamma + (\gamma + \beta^2)\beta - 5z^2\beta = a(C - z)(C - 2z)a^*$. Hence

$$R_5 = \gamma(\beta\gamma + (\gamma + \beta^2)\beta - 5z^2\beta) = a(BH - 2z^2)(C - z)(C - 2z)a^* = -aBC(C - z)(C - 2z)a^*,$$

which vanishes by (B.3). For R_4 , put $g = \gamma + \beta^2$, so that $R_4 = (g - z^2)(g - 4z^2) + \beta\gamma\beta$. The same rules give

$$\begin{aligned} (g - z^2)(g - 4z^2) &= 18z^4 + a(27z^3 - 9z^2C + (3z - C)H(3z - C))a^*, \\ (3z - C)H(3z - C) &= -27z^3 - 9z^2B + 11z^2C + 3zBC + 3zCB - CBC, \\ \beta\gamma\beta &= -18z^4 + a(9z^2B - 2z^2C - 3zBC - 3zCB + CBC)a^*, \end{aligned}$$

where the second line uses $C^3 = 3zC^2 - 2z^2C$. The sum is zero. $\square$

The program `tools/check_e8_rho_relations_formal.py` repeats these reductions mechanically over $\mathbb{Z}[z]$, without specializing z .

Remark B.4 (Sign conventions in [17, Example 4.26]). Lemma B.3 agrees with the last two relations of Karmazyn’s intermediate presentation, written in the invariants R_1, R_2, R_3, S_2 of the five-arm parameters $\tau_1, \dots, \tau_5$. On the ρ -line, $(\tau_1, \dots, \tau_5) = (2, 1, 0, -1, -2)z$ and $(R_1, R_2, R_3, S_2) = (3z, 2z^2, 0, 2z^2)$; here $\beta = a_4 a_4^* + R_1$ and $\gamma = a_4 B a_4^* - S_2$. For the quadratic and cubic relations displayed there, the shifts must be $B = b + \tau_5$ and $C = c + \tau_3$. The final display of [17, Example 4.26] rewrites these relations through $T_3 = R_2 + S_2 + R_1 S_1$, $T_2 = R_3 + R_2 S_1 + R_1 S_2$, $T_1 = -R_3 R_1 + R_2 S_2$ and $T_0 = R_3 S_2$. These are $+\sigma_2, \dots, +\sigma_5$ of the τ_i , whereas the list of invariant generators there sets $T_i = -\sigma(\tau)$. Substituting the latter convention into the final display reverses the signs of the parameter terms and gives

$$R'_4 = (\gamma + \beta^2)^2 + \beta\gamma\beta + 5z^2(\gamma + \beta^2) - 4z^4, \quad R'_5 = \gamma\beta\gamma + \gamma^2\beta + \gamma\beta^3 + 5z^2\gamma\beta.$$

These are not relations of B_5 . In degree four, the relation rows of Q_ρ are $[P]$ and $[R_4]$. By Section 3.6.1, the surjection $HH_0(Q_\rho)_4 \rightarrow HH_0(B_5)_4$ is an isomorphism, so the classes of degree four that vanish in $HH_0(B_5)$ are exactly the span of $[P]$ and $[R_4]$. In cyclic words,

$$\begin{aligned} [P] &= [\beta^4] - 22z[\beta^3] + 179z^2[\beta^2] - 638z^3[\beta] + 840z^4, \\ [R_4] &= [\gamma^2] + 3[\beta^2\gamma] + [\beta^4] - 5z^2[\gamma] - 5z^2[\beta^2] + 4z^4, \end{aligned}$$

and $[R'_4] - [R_4] = 10z^2[\gamma] + 10z^2[\beta^2] - 8z^4$. If $[R'_4]$ were in the span, comparing the coefficients of $[\gamma^2]$ and of $z[\beta^3]$ would force $[R'_4] = [R_4]$, which is false. Hence $R'_4 \neq 0$ in B_5 .

An earlier rank calculation, archived in the companion repository at commit [5a2001f](#) (the archived E_8 checker sources, the recorded E_8 outputs, and the E_8 computation and reproduction guide), used R'_4 and R'_5 . Its rank table coincides with the one below, but it is not used in this thesis.

B.2 Relation matrices and exact ranks

The C++17 program `tools/check_e8_corrected_rho.cpp` enumerates, in each half-path degree $m \leq 23$, the cyclic-word columns $z^a[w]_{\text{cyc}}$ and the rows $[Rw]_{\text{cyc}}$ for $R \in \{P, R_4, R_5\}$ and words w . It performs Gaussian elimination over $\mathbb{F}_{1000003}$ in exact modular arithmetic. The command `make math` compiles it, repeats the calculation over $\mathbb{F}_{1000033}$ with only the modulus changed, and compares both outputs with the table below. Both primes give the same ranks. One prime suffices for Proposition 3.14; the second is an independent check. The table lists the degree, the column count c_m , the modular rank r_m , and the characteristic-zero nullity bound $u_m = c_m - r_m$.

Degree m	Columns c_m	Rank r_m over $\mathbb{F}_{1000003}, \mathbb{F}_{1000033}$	Nullity bound u_m
0	1	0	1
1	2	0	2
2	4	0	4
3	6	0	6
4	9	2	7
5	12	5	7
6	17	10	7
7	22	16	6
8	30	24	6
9	40	34	6
10	55	49	6
11	74	69	5
12	105	100	5
13	146	142	4
14	210	206	4
15	304	300	4
16	447	443	4
17	658	655	3
18	987	984	3
19	1480	1478	2
20	2246	2244	2
21	3416	3414	2
22	5227	5225	2
23	8014	8013	1

The SHA-256 checksums of the two E_8 programs are:

File	SHA-256
<code>check_e8_corrected_rho.cpp</code>	2545d84e337ea8b4ac650c428b90880a36ab47cd168e8935e0d4598ac1dbabd9
<code>check_e8_rho_relations_formal.py</code>	208305e7e5a7613b75b66959b4ac7022c31644b54c988cbd642baad7382d5f1a

C Exact linear algebra for $D_{2,2}$

This appendix describes the computations behind Proposition 4.19 and the check of the condition (4.8) used in Proposition 4.20. They use exact rational arithmetic only (`fractions.Fraction` in Python), without floating-point approximation, modular reduction, or a computer algebra system. The programs are `tools/verify_conjecture_1_4_counterexample.py` and `tools/check_d22_tangent_image.py`; the command `make math` runs both. Neither program is needed for Theorem 4.16 or Corollary 4.17, whose proofs are symbolic and valid for all $n \geq 2$.

Cochains and ranks. The algebra $A = D_{2,2}$ is given in the basis $(1, x, x^2, x^3, x^4, y, xy, x^2y)$ by the multiplication table of Lemma 4.14(i), and the program checks associativity and the unit on all triples of basis vectors. The centre Z_2 has the basis $(1, u, v, w, q)$, embedded as $(1, x^2, x^3, x^2y, x^4)$, and the program checks that the embedding is multiplicative. Normalized cochains are multilinear maps on the span of the basis vectors other than 1; for Z_2 , the second cochains are symmetric. The coboundary maps are sparse rational matrices, and exact row reduction gives the following dimensions.

	A , associative	Z_2 , commutative
normalized 1-cochains	56	20
rank of δ on 1-cochains	50	10
normalized 2-cochains	392	50 (symmetric)
rank of δ on 2-cochains	339	24
2-cocycles	53	26
cocycles modulo coboundaries	$HH^2(A)$: 3	$T_{\text{com}}^1(Z_2)$: 16

The derivations of A form the kernel on 1-cochains, of dimension $56 - 50 = 6$; inner derivations have dimension $8 - 5 = 3$, so $\dim HH^1(A) = 3$.

The centre tangent map. For each of the 53 basis cocycles ϕ of $Z_{\text{norm}}^2(A, A)$, the first program solves the linear system (4.2) for τ with values in the complement $\text{span}(x, y, xy)$ of Z_2 . It checks that the solution exists and is unique, forms ψ by (4.3), and checks that ψ is symmetric, takes values in Z_2 , and is a cocycle. It also verifies that the coboundaries of A map to commutative coboundaries and that the 53 cocycles ψ have rank 3 modulo $B_{\text{com}}^2(Z_2)$. Finally, the coefficient of w in $\psi(v, v)$ vanishes on all these cocycles and on $B_{\text{com}}^2(Z_2)$, but equals 1 on ψ_0 . This gives a second, computational proof of Corollary 4.17(i) for $n = 2$.

Representatives and the socle condition. The second program checks that ψ_1, ψ_2, ψ_3 of Proposition 4.19(iii) are commutative cocycles whose classes lie in the image of χ_A and span it. Reduce them modulo $B_{\text{com}}^2(Z_2)$ by the reduced row-echelon basis of the coboundaries. Their coefficients of w in $\psi(u, v)$, of w in $\psi(u, w)$, and of q in $\psi(u, q)$ are non-pivot coordinates. They form the matrix

$$\begin{pmatrix} 0 & -1 & -2 \\ -\frac{1}{2} & 0 & 0 \\ 0 & -2 & -2 \end{pmatrix}, \quad \det = -1,$$

whose rows correspond to ψ_1, ψ_2, ψ_3 . The program then verifies (4.8) for all 53 image cocycles and for the 20 generators of $B_{\text{com}}^2(Z_2)$, as a polynomial identity in the coordinates of $s = \alpha v + \beta w + \gamma q$, and checks that ψ_0 violates it.

Uniform checks. For $n = 2, \dots, 6$, the first program also checks the identities of Lemma 4.14 that enter the proof of Theorem 4.16: $\dim A_n = 4n$, $\dim Z(A_n) = n + 3$, the centralizer of x has dimension $2n + 2$, $wx = xw = 0$, and $vA_n + A_nv \subseteq \mathbb{C}[x]$. It also checks that the multiplication of (4.6) for $n = 2$ is associative over $\mathbb{Q}[t]$. These finite checks illustrate, but do not replace, the proofs for general n .

The SHA-256 checksums of the two programs are:

File	SHA-256
<code>verify_conjecture_1_4_counterexample.py</code>	<code>e51dd6b0816f019f3cd2f38c9fb12312ec4b9f49d18323a18748055bdbb7cce</code>
<code>check_d22_tangent_image.py</code>	<code>2140afdf251c2fb1f123a02cdb92fac87b4d029e822f336a4a2ca1e9e92dc9ae</code>

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